Online Appendix to

The Transmission of Financial Stress to the U.S. Real Economy: A BVAR Analysis with Non-Gaussian Identification

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This appendix collects the material left out of the paper: the construction of the two panels (Section A), the derivations and estimation detail behind the identification (Section B) and behind the bridge to monthly frequency (Section C), the formal and empirical comparison of the two contractions of the cumulant tensor (Section D), the question of whether the fourth moment exists (Section E), and the additional tables, figures and robustness checks referenced from the paper (Section F). Section and equation numbers here are the ones the paper points to. Notation follows the paper throughout.

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A Data

A.1 Sources and identifiers

Macroeconomic and fixed-income series come from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. Equity indices come from the LSEG Datastream platform, which is preferred where both providers carry a series because its historical coverage is longer. Table A.1 lists every series with the identifier needed to reproduce the download.

Table A.1: Source identifiers. FRED mnemonics are as published by the Federal Reserve Bank of St. Louis; LSEG entries are the worksheet names in the Datastream export. BEX denotes the Bekaert–Engstrom–Xu indices and GZ the Gilchrist–Zakrajšek excess bond premium.

Variable	Source	Identifier	Transformation
Daily panel			
SP500	LSEG	`S&P500 daily`	$\ln(\cdot)$
VIX	FRED	`VIXCLS`	$\ln(\cdot)$
Baa	FRED	`DBAA`	level (%)
Aaa	FRED	`DAAA`	level (%), spliced at 2001-12-07
CSPREAD	—	Baa − Aaa	level (pp)
Monthly panel			
INDPRO	FRED	`INDPRO`	$\ln(\cdot)$
UNRATE	FRED	`UNRATE`	level (%)
HOUST	FRED	`HOUST`	$\ln(\cdot)$
CSHPI	FRED	`CSUSHPISA`	$\ln(\cdot)$
PCEPI	FRED	`PCEPI`	$\ln(\cdot)$
NRCREDIT	FRED	`NONREVSL`	$\ln(\cdot)$
FFR	FRED	`FEDFUNDS`	level (%)
RAI_BEX	BEX	risk aversion index	level
UNC_BEX	BEX	uncertainty, ann. vol %	level
EBP	GZ	excess bond premium	level (pp)

The credit variable is the difference between the two Moody's seasoned yield series,

$$\text{CSPREAD}_t \;=\; \text{Baa}_t - \text{Aaa}_t, \tag{A.1}$$

which isolates compensation for default and downgrade risk within the corporate segment, net of the term structure. Entering the two levels separately would introduce a near-collinear pair driven by a common risk-free factor. In the extended seven-variable system, where the Aaa level enters on its own, the differential also separates a generalised

repricing of corporate yields from a widening of the quality premium on lower-grade credit.

One property of CSPREAD should be recorded, because it returns in Section F.1. A spread is not automatically free of serial correlation. CSPREAD displays the largest residual Breusch–Godfrey statistic in the system at every lag order considered, which reflects genuine slow-decaying persistence together with conditional heteroskedasticity, and extending the lag order does not remedy it.

A.2 The Aaa splice

The published Moody's seasoned Aaa series changes composition on 7 December 2001, and the two segments are joined by an additive level adjustment of +0.0048 percentage points applied to the earlier one. The adjustment is a single constant over the whole pre-splice sample, so it shifts the level of the Aaa series and of CSPREAD uniformly and leaves every daily change unaffected except at the join itself, where it contributes 0.23 standard deviations of ΔCSPREAD on one day. It is therefore inert for an identification operating on the fourth moments of VAR residuals: a location shift does not enter a centred cumulant, and a single sub-one-sigma increment cannot influence a fourth moment estimated on 9,058 observations.

The point is worth recording because of one coincidence in the narrative table of the paper. The tenth-largest realisation of the credit shock falls on 11 December 2001, four trading days after the join. It is driven by a 7.1 standard-deviation move in the spread itself and not by the splice, whose effect had already passed through the series on 7 December and was in any case two orders of magnitude smaller.

A.3 The VIX in levels: numerical record

Section 2 of the paper argues the logarithmic transformation of the volatility index on distributional and identification grounds. This section records the numerical comparison behind the claim that the raw index also degrades the estimation mechanically.

The two episodes that dominate the fourth moment of the raw index are worth naming, since the argument turns on what they are. On 5 February 2018 the collapse of the inverse-volatility exchange-traded

Table A.2: Raw versus log VIX. Skewness and excess kurtosis are of the variable in levels. Condition numbers are of the posterior mean of Σ_u at the selected lag order. Adjacent eigenvalue pairs are counted in the extended system, which has six of them and is therefore the specification in which degeneracy can be counted at all.

	Raw VIX	log VIX
Skewness (levels)	2.20	0.67
Excess kurtosis (levels)	8.63	0.49
$\kappa(\widehat{\Sigma})$, extended system $M = 7$	1.12×10^5	348.7
$\kappa(\widehat{\Sigma})$, baseline system $M = 3$	—	75.0
Near-degenerate adjacent pairs (of six)	5	2
Component carrying the stress signature	v_7 (last)	v_1 (first)

products (XIV, SVXY) forced a mechanical rebalancing of short VIX-futures positions into the close; on 27 January 2021 a retail short squeeze spilled into index volatility through dealer gamma hedging. On both dates the equity drawdown was modest and quickly reversed, and the Baa−Aaa spread was essentially unchanged. In levels their weight in the kurtosis criterion is mechanical and not economic: a forty per cent rise in the index is +6 points from VIX = 15 and +20 points from VIX = 50, and at the fourth power a handful of such residuals can decide a ranking. Under the logarithm the two dates remain the largest realisations of the third component, at 12.53σ and 7.58σ, but that component is the least kurtotic of the three, which is the separation the transformation buys.

Why the conditioning matters more here than it would in a second-moment exercise is the substance of Table A.2. The identification whitens the residuals as $\boldsymbol{\iota}_t = \widehat{\Sigma}^{-1/2}\hat{u}_t$ and builds $\widehat{K}$ from their *fourth* moments, so an ill-conditioned $\widehat{\Sigma}$ makes $\widehat{\Sigma}^{-1/2}$ fragile and any whitening error is amplified quartically. The damage is visible directly: under the raw index the whitening check fails its tolerance, with $\max|\mathrm{Cov}(\boldsymbol{\iota}) - I_M| = 1.28 \times 10^{-3}$ against a target of 10^{-3}, where the logarithm passes comfortably at 5.51×10^{-4}. The condition number is the informative statistic here because it is dimensionless. The posterior draw-variance of Σ_u also falls by five orders of magnitude between the two specifications, but the elements of Σ scale with the fourth power of the units of the data, so a change of that size is what a change of scale mechanically produces and is evidence of nothing.

A.4 Sample, transformations and the October 2025 gap

The raw monthly panel covers 432 months, January 1990 to December 2025, overlapping the monthly-aggregated daily shocks over 431 of them. The estimation sample is nonetheless truncated at September 2025, giving 429 months and 414 usable observations after the 15 lags of Section C.4.

The federal government shutdown of 1 October to 12 November 2025 interrupted the collection programme of the Bureau of Labor Statistics. The unemployment rate, one of the ten series, is permanently, not merely belatedly, missing for October 2025, because the household survey was never fielded; the October CPI, which is collected in the field during the reference month and is an input to the PCE price index, was never collected either. Series produced by the Federal Reserve are unaffected.

Dropping the incomplete row would leave 2025-09 followed immediately by 2025-11. Because VAR lags are built by row position and not by date, such a hole silently pairs y_t with the wrong $y_{t-1}, \ldots, y_{t-p}$ for the p observations that straddle it, and does so at the end of the sample, where the instrument is most informative. The estimation code carries an explicit calendar-contiguity guard against this. Ending the sample before the gap costs two months, November and December 2025, and we prefer that to interpolating the unemployment rate and the price index across a month of acute fiscal disruption.

A.5 Calendar alignment

Raw daily series from FRED and LSEG are published on heterogeneous schedules. To ensure temporal consistency the daily dataset is aligned to the New York Stock Exchange trading calendar in two steps: the raw series are reindexed onto NYSE trading days using the `pandas_market_calendars` library, and the resulting panel is forward-filled so that FRED series not published on a given trading day inherit their most recent observation. No forward-filling occurs across weekends or holidays, and the per-variable audit of how many cells this fills is produced by the alignment script and reported in its log. The resulting daily dataset contains $T = 9{,}067$ NYSE trading days spanning January 1990 to 2025. The daily-to-monthly bridge required by the second

stage is not a calendar alignment problem but a temporal-aggregation problem, and is treated as such in Section C.2.

A.6 Estimation in levels

Both VARs are estimated on data in levels, without preliminary stationarity transformations. For the daily system the justification is Sims et al. (1990), who show that OLS coefficient estimates remain consistent even when variables contain unit roots, provided no restrictions are imposed on the cointegrating rank; under the flat prior adopted there the argument applies directly. Estimation in levels preserves the information in long-run co-movements, avoids pre-specifying cointegrating relationships, and is consistent with the random-walk behaviour of daily financial data.

The Sims et al. (1990) result concerns the conditional mean, whereas what the identification consumes are the fourth moments of the residuals, for which it offers no analogue. That gap is closed empirically rather than by appeal: re-running the identification on the first-differenced system moves the rotation columns by about one degree and leaves the eigenvalue ordering and the anchor variables unchanged (Section F.3).

The monthly system rests on a different argument, because its prior is informative and the Sims et al. (1990) result does not transfer to it. Estimation uses the hierarchical Minnesota prior of Giannone et al. (2015), in which the shrinkage hyper-parameters are estimated rather than fixed. The mixed $I(0)/I(1)$ structure is handled in two ways. The prior mean is set variable by variable, with the seven trending series centred on a random walk ($\delta_i = 1$) and the three mean-reverting risk premia centred on white noise ($\delta_i = 0$). And the sum-of-coefficients and dummy-initial-observation priors, which govern how much unit-root and co-persistence behaviour the prior imposes, are chosen by maximising the marginal likelihood jointly with the overall tightness, so the degree of shrinkage towards the unit-root representation is determined by the data. Overall tightness is initialised by a preliminary grid search over $\tau \in \{0.1, \ldots, 50\}$, and the optimised hyper-parameters are carried unchanged into the augmented VAR so that the two specifications share the same prior.

A.7 The financial block of the monthly panel

The last three variables of the monthly panel deserve separate discussion, because their role in the design is easy to misread. They are ordinary control variables in the VAR. They are not instruments, not anchors, and no shock is matched to any of them by assumption; the identification is agnostic and nothing in it uses the names of these series.

- **EBP** is the Gilchrist–Zakrajšek excess bond premium (Gilchrist and Zakrajšek, 2012): the component of the corporate credit spread not accounted for by measured default risk, obtained by projecting bond-level spreads on issuer-specific default probabilities and bond characteristics and averaging the residual. It is a purified measure of the price of bearing credit risk, stripped of expected-default compensation.

- **RAI_BEX** is the Bekaert–Engstrom–Xu risk aversion index (Bekaert et al., 2022): the component of asset-price variation attributable to time-varying preferences rather than to the quantity of risk, extracted from a dynamic no-arbitrage model estimated on asset prices and macroeconomic data.

- **UNC_BEX** is the companion Bekaert–Engstrom–Xu uncertainty index, in annualised volatility percentage points, which isolates the quantity of risk.

The block is indispensable rather than decorative on two counts, both established empirically in the paper. It is the transmission channel. A financial shock does not move industrial production within the calendar month in which it occurs, production being a physical flow already largely realised, and the first-stage diagnostics confirm this starkly. The shock enters the system through financial prices and reaches the real economy through the estimated dynamics, so removing this block would leave the shock with nothing to load on at impact and would destroy the identification instead of purifying it. And EBP supplies the unit of measurement. Because the identification returns shocks of arbitrary scale, the impulse responses must be normalised before they can be read, and Section C.6 normalises every shock to a common movement of the excess bond premium on impact. Using a common reference is what keeps the three impulse-response panels on one yardstick; using

a different reference for each shock would require deciding in advance which shock is the credit one, which would reintroduce through the back door the labelling the design is built to avoid.

A.8 External validation series

A separate group of monthly series is used only in the external validation and enters no estimation step: not the daily VAR, not the identification, not the monthly macro panel. Keeping them outside the estimation is what gives the exercise its force, since a correlation that appears there cannot have been engineered by the specification.

Two families are used. The news-based Economic Policy Uncertainty indices of Baker et al. (2016) are constructed from the frequency of newspaper articles jointly mentioning economic policy and uncertainty: the US series (US_EPU) and the two global aggregates of Davis (2016) (GEPU_current and GEPU_ppp, differing in whether national indices are combined with current-GDP or purchasing-power-parity weights). The Geopolitical Risk indices of Caldara and Iacoviello (2022) are built analogously from newspaper coverage of geopolitical tensions, in a global (GPR) and a US-specific (GPRC_USA) variant. The GPR family serves as a discriminating placebo: it measures a genuine source of uncertainty that has no particular reason to drive the amplitude of US financial-market innovations, so a design reporting significant associations for it as readily as for EPU would be diagnosing spurious common trending rather than economic content.

Coverage differs across indices, with US_EPU beginning in 1985 and GEPU in 1997, so each correlation is computed on the intersection of the index's own support with the monthly shock sample, and the effective window is reported alongside every coefficient instead of imposing a common truncated sample on all five.

B Identification: derivations and estimation detail

B.1 The case for a flat prior

The daily VAR is estimated under a flat prior, with the lag order selected on the data by marginal likelihood over $p \in \{3, \ldots, 10\}$. The choice is one of minimal commitment: an identification that imposes no sign, zero or timing restriction should not reintroduce structure through the prior, and under a flat prior the estimation of $\{\Phi_j\}$ reduces to equation-by-equation OLS.

Little is given up. Identification operates entirely through $\bar{\Sigma}$ and the residuals it whitens, and for large T the posterior mean of Σ_u converges to the OLS estimate whatever the prior on $\{\Phi_j\}$, so $\bar{\Sigma} \to S_{\text{OLS}}$; at $T = 9{,}058$ the weight on S_{OLS} is effectively one. The point can be checked rather than assumed. Re-estimating the system at a common lag order under the GLP-strict Minnesota prior of Giannone et al. (2015) leaves the pre-whitened residuals indistinguishable from their flat-prior counterparts, with the two series correlating at 0.99998 or above in every equation, and their excess kurtosis, the only feature of the residuals the identification consumes, agreeing to two decimal places. Table B.1 reports how little the prior moves the conditional mean.

Table B.1: How much the Minnesota prior moves the estimate. Own first-lag coefficient under the flat prior and under the Giannone et al. (2015) Minnesota prior at $p = 22$, with shrink $= (\phi^{\text{MN}} - \phi^{\text{OLS}})/(1 - \phi^{\text{OLS}})$: zero means the data dominate, one means the prior does.

Variable	Flat prior	Minnesota (GLP)	Prior mean	Shrinkage fraction
SP500	0.885	0.891	1	0.054
VIX	0.907	0.911	1	0.043
CSPREAD	0.852	0.850	1	−0.018

Even with hyperparameters optimised by the sequential empirical-Bayes procedure of Giannone et al. (2015), the prior moves the own first-lag coefficients by four to five per cent of the distance to its own mean, and in one case moves them away from it.

The flat prior is then favoured on two counts, neither of which is that the alternative fails. The first is that it keeps the lag order a

decision of the data. Under a Minnesota prior the shrinkage absorbs superfluous lags, so p ceases to be sharply determined and must be fixed on other grounds, which is why the Minnesota specification reported for comparison sets $p = 22$ a priori, at one trading month, instead of selecting it. Under a flat prior nothing absorbs an unnecessary lag, so the marginal likelihood discriminates: extending the candidate grid to $p \in \{9, 12, 15, 18, 22, 25\}$ leaves it monotonically decreasing beyond the selected order, so the choice is interior to the data and not to the prior.

The second is a matter of internal consistency, though a narrower one than it first appears. Both priors sit inside the same Gaussian likelihood, since the flat-prior posterior is Normal–Inverse-Wishart, derived under $u_t \sim \mathcal{N}(0, \Sigma_u)$ exactly as the GLP marginal likelihood is, so neither escapes the tension between a Gaussian estimation stage and an identification that exploits the failure of Gaussianity. What differs is the role the assumption plays. Under the flat prior it is a working device whose output survives its own misspecification, since the posterior mean of $\{\Phi_j\}$ is OLS and OLS is consistent under general innovations (Sims et al., 1990). Under GLP the quantity used is the *argmax* of the Gaussian criterion, and no analogous result guarantees that the selected degree of shrinkage remains appropriate when the likelihood is misspecified in the direction the identification relies on. Where Gaussianity does bind under the flat prior, in the width of the posterior bands, which a Normal–Inverse-Wishart posterior understates under heavy tails, the results do not rely on it, since reported uncertainty is bootstrap-based throughout.

The monthly macro VAR is a different case. It is not identified through higher moments, its sample is two orders of magnitude shorter (429 against 9,058 observations), and it contains mixed integration orders, so a Minnesota prior is appropriate and is used (Section C.4).

B.2 Lag selection

With the flat prior, the log-marginal likelihood $\ln p(\boldsymbol{Y} \mid p)$ is available analytically for each candidate order in the grid $\mathcal{P} = \{3, \ldots, 10\}$, and the selected order is

$$p^* = \arg\max_{p \in \mathcal{P}} \ln p(\boldsymbol{Y} \mid p, \text{ flat prior}). \tag{B.1}$$

The marginal likelihood penalises complexity through the Occam's razor effect, since probability mass is spread more thinly over larger parameter spaces, so additional lags must be strongly supported by the data. It is preferred to AIC or BIC because it provides a coherent Bayesian model comparison that accounts for the full posterior instead of a point estimate. The realised surface is in Table F.1.

B.3 Residual diagnostics

Serial correlation. For each equation the Breusch–Godfrey LM test (Breusch, 1978; Godfrey, 1978) is applied with $H = 10$ auxiliary lags, yielding

$$\mathrm{LM}_{\mathrm{BG}} = (T_{\mathrm{eff}} - H) \cdot R^2_{\mathrm{aux}} \xrightarrow{d} \chi^2(H) \tag{B.2}$$

under the null of no serial correlation up to order H. One caveat is central to the interpretation: the test is derived under conditional homoskedasticity, and in the presence of strong GARCH effects the statistic is inflated and over-rejects even when the conditional mean is correctly specified. In daily financial data a rejection must therefore be read as a joint signal of possibly minor conditional-mean autocorrelation and typically dominant conditional heteroskedasticity.

Non-Gaussianity. Non-Gaussianity is assessed on the pre-whitened residuals $\boldsymbol{\iota}_t = \widehat{\Sigma}^{-1/2}\hat{u}_t$, which have identity covariance by construction, through three complementary measures. The sample excess kurtosis is $\hat{\kappa}_j = T^{-1}\sum_t \hat{\iota}^4_{j,t} - 3$. The Hill tail index (Hill, 1975) uses the upper m order statistics of the absolute residuals,

$$\hat{\alpha}_j = \left[\frac{1}{m}\sum_{i=1}^{m} \ln|\hat{\iota}_{(i),j}| - \ln|\hat{\iota}_{(m+1),j}|\right]^{-1}, \tag{B.3}$$

with $m = \lfloor 0.05 \cdot T_{\mathrm{eff}} \rfloor$. Under a power-law tail $\Pr(|\iota_j| > x) \sim x^{-\alpha}$, a value $\hat{\alpha}_j < 4$ implies a potentially infinite fourth moment in that tail. That reading must be qualified, and Section E does so: the Hill estimator presupposes a Pareto tail and returns a finite index below 4 even for mixtures whose moments are all finite. The statistic is therefore reported as a description of tail thickness and not as a verdict on moment existence. Either way, a value well below the Gaussian benchmark for at least $M - 1$ components establishes the non-Gaussianity the identification requires. What heavy tails threaten

is not the consistency of $\widehat{K}$ but its sampling variance, which is why inference on the eigenvalues rests on the block bootstrap of Section B.8 and not on a regularisation of $\widehat{K}$ itself.

The Kim–White robust excess kurtosis is the octile-based measure of Moors (1988) (KR_2 in Kim and White, 2004),

$$\mathrm{KW}_j = \frac{(E_7 - E_5) + (E_3 - E_1)}{E_6 - E_2} - 1.233, \qquad E_k = F_j^{-1}(k/8), \quad \text{(B.4)}$$

where 1.233 is its Gaussian value. It depends only on octiles, so it is unaffected by the extreme observations that dominate the sample kurtosis, and it is on a different scale: a Laplace distribution gives 0.35 and a Student-t with five degrees of freedom 0.09. Positive values confirm non-Gaussianity in the bulk of the distribution and not in the extreme tails alone, which matters here: with Hill indices below 4, one must rule out the possibility that the measured kurtosis is an artefact of a few extreme days.

B.4 Implementation and the ridge

Fourth moments of heavy-tailed data are estimated imprecisely, and a small number of extreme observations can dominate the sample fourth-moment matrix $\widehat{K}$. The estimator therefore admits a ridge shrinkage of $\widehat{K}$ toward its Gaussian counterpart, applied before the excess is taken,

$$\widehat{K}_{\mathrm{reg}} = w\,\widehat{K} + (1 - w)\,K_z, \qquad w = \frac{1}{1 + \tau}, \qquad \text{(B.5)}$$

governed by a single scalar $\tau \geq 0$. The device is benign for identification in a precise sense. Because K_z is invariant under orthogonal conjugation, the regularised excess is $\widehat{K}_{\mathrm{reg}} - K_z = w\,(\widehat{K} - K_z)$, an exact scalar multiple of the unregularised one. Every step that follows is either linear in that matrix or invariant to its scale, so the shrinkage divides each eigenvalue by $1 + \tau$ and leaves the eigenvectors, the ordering, the eigenvalue ratios, the degeneracy probabilities and every downstream object numerically untouched.

The baseline sets $\tau = 0$, so $w = 1$ and no shrinkage is applied, for two reasons. Nothing is bought by it: since the effect is an exact common rescaling, it cannot stabilise a direction or a ranking, only the units in which the magnitudes are read. And something is lost. Under

Assumption 1 of the paper the eigenvalues are estimates of the excess kurtoses κ_i, and shrinking them biases those estimates towards zero by a known factor, contaminating exactly the comparison between d_i and κ_i that Section D.5 uses as a specification check. At $\tau = 0$ the eigenvalues reported are raw fourth-moment estimates. The imprecision that motivates a ridge is real, but it is a statement about sampling variance, and it is priced where it belongs, in the block bootstrap of Section B.8, and not by a deterministic rescaling of the point estimate. Setting $\tau = 0.05$ instead would divide every eigenvalue reported in this project by 1.05 and change nothing else.

The identical chain from $\widehat{K}$ to the eigendecomposition is applied at three points: to the posterior mean of Σ_u, to construct the reference rotation; to each of the 5,000 posterior draws; and to each moving-block bootstrap resample. It is implemented once as a single routine so that the three cannot drift apart.

B.5 Sampling frequency: the aggregation argument

The identification extracts the rotation from the fourth-moment structure of u_t, so how much identifying information is available is not a fixed property of the three series: it depends on the frequency at which they are sampled. That makes the choice of frequency part of the identification design.

Write $u^{(n)}$ for the sum of n consecutive innovations, the operation that turns a daily series into a weekly, monthly or quarterly one. For independent and identically distributed components, fourth cumulants and variances are both additive, so

$$\kappa\big(u^{(n)}\big) = \frac{\operatorname{cum}_4\big(u^{(n)}\big)}{\operatorname{Var}\big(u^{(n)}\big)^2} = \frac{n \operatorname{cum}_4(u)}{n^2 \operatorname{Var}(u)^2} = \frac{\kappa(u)}{n}. \tag{B.6}$$

Aggregation is a contraction on precisely the object the estimator consumes: the central limit theorem seen through the fourth cumulant.

Equation (B.6) is a benchmark and not a description of financial data. Volatility clusters, so extreme days arrive together and their sums stay extreme; under GARCH-type dependence the excess kurtosis of the aggregate still declines towards zero, but far more slowly than $1/n$

(Drost and Nijman, 1993). The residuals of the daily system make the gap concrete. Aggregating them over 21 trading days leaves an excess kurtosis of 6.90 for the S&P 500 and 10.58 for the credit spread, against the 0.48 and 1.65 that independence would predict, so the credit spread retains roughly 30 per cent of its daily excess kurtosis where (B.6) predicts five.

What deteriorates decisively is not the level of the fourth moment but the precision with which the spectrum can be resolved. Identification rests on the separation between adjacent eigenvalues of the kurtosis matrix, which is what the near-degeneracy test measures. Aggregating by n acts on that separation twice over: it compresses the eigenvalues towards one another, and it divides the sample by n, so the standard error of each estimated eigenvalue grows as the gap it must resolve shrinks. A common contraction of all κ_r would in principle leave their ratios, and hence population identification, intact; what is lost is the ability to distinguish them in a finite sample. The same daily residuals aggregated to quarterly frequency leave 143 observations, at which point the estimated ordering of the components is no longer stable. This, and not the absence of non-Gaussianity, is why higher-moment identification has found its most convincing applications in high-frequency financial data.

The two ingredients are therefore complementary and not merely compatible: higher-order moments supply the identification device, and high frequency supplies both the non-Gaussianity the device exploits and the sample size it needs to exploit it. In the present data the pre-whitened daily innovations have excess kurtoses between 9 and 35 over 9,058 observations.

B.6 Conditional heteroskedasticity

Lanne et al. (2017) show, in an extension of their main result, that identification survives conditionally heteroskedastic shocks, GARCH dynamics among them, provided the volatility processes are independent across shocks. The eigen-estimator of the rotation operates on the unconditional fourth moment, whose sample counterpart converges under stationarity and ergodicity even when the shocks are not i.i.d.: serial dependence does not bias the point estimate, it inflates the sampling variance of the kurtosis matrix. A volatility factor *common* to

all shocks falls outside that extension, since it makes the shocks dependent; Lemma B.1 below shows that the Móri contraction is nonetheless robust to it.

This distinction dictates the treatment adopted. A natural-looking response to volatility clustering would be to pre-filter the residuals through a GARCH model and identify on the standardised innovations. That route is not taken, and deliberately so. Standardising by a fitted conditional scale removes exactly the feature the identification exploits: the large realisations that generate the fourth-moment signal are, by construction, the ones a conditional-variance model attributes to high fitted volatility, so the filtered series is closer to Gaussian than the object of interest and the eigenvalue spectrum it produces is an artefact of the filter. The filter would moreover be fitted equation by equation, whereas the identifying object is the joint tensor of the pre-whitened system, so a univariate rescaling has no reason to preserve the contemporaneous mixing that the rotation is meant to recover. Since neither serial dependence (Lanne et al., 2017) nor a common volatility factor (Lemma B.1) biases the rotation in the first place, filtering would forfeit signal in exchange for a bias correction that is not needed.

The consequence of dependence is therefore accepted and quantified instead of suppressed: the inflated sampling variance of $\widehat{K}$ is measured directly by the moving-block bootstrap of Section B.8, and the assumption that actually matters, contemporaneous independence, is tested on the identified shocks with statistics chosen to be robust to common volatility.

How exact is independence, and what the sample size buys. The independence assumption is exact rather than approximate in what it asks of the fourth moments, since independence is equivalent to diagonality of the cumulant tensor in the structural basis, and the estimator delivers that diagonality by construction rather than discovering it. The data are never asked *whether* the components are independent, only in which *direction* the independence is to be imposed, and that direction is estimated from sample cross-cumulants. Sample size therefore binds on a margin distinct from the eigenvalue separation of Section B.5: the sampling variance of a fourth cumulant involves an eighth moment, so under heavy tails the cross-cumulants the

rotation is meant to annihilate can themselves be indistinguishable from noise, and the estimator then buys exact diagonality by rotating onto whichever direction the sampling error happens to favour. The imposed independence would still be real, but it would be the independence of an artefact, and nothing would signal it, since the routine always returns a rotation and that rotation always diagonalises the sample tensor.

What the daily sample buys is not exactness but the ability to *locate* the residual departure, and two readings taken by different routes agree on where it sits. Nine of the twelve SV-robust diagnostics accept, with the three rejections falling on the single pair v_2–v_3. The specification check of Section D.5 does not bear on this question: by Lemma B.1 the gap between the Móri eigenvalues and the realised kurtoses is exactly the excess co-kurtosis, which measures the common volatility factor rather than a failure of independence. The assumption binds approximately, and at this sample size the departure is small enough, and located enough, to be attributed instead of merely tolerated.

Common volatility and the Móri contraction.

Lemma B.1. *Let the whitened structural shocks satisfy* $v_t = \omega_t^{1/2} z_t$, *where* $\omega_t > 0$ *with* $\mathbb{E}[\omega_t] = 1$, *the components of* z_t *are mutually independent with* $\mathbb{E}[z_{i,t}] = 0$ *and* $\mathbb{E}[z_{i,t}^2] = 1$, *and* z_t *is independent of* ω_t. *Then the Móri matrix* $K(Z) = \mathbb{E}[(v_t'v_t)\, v_t v_t'] - (M+2)I_M$ *is diagonal in the structural basis, with diagonal entries*

$$d_i = \kappa_i + \sum_{j \neq i} \big(\mathbb{E}[v_i^2 v_j^2] - 1\big), \qquad \kappa_i = \mathbb{E}[v_i^4] - 3.$$

Proof. For $p \neq q$, $K_{pq} = \mathbb{E}[\omega_t^2] \sum_i \mathbb{E}[z_i^2 z_p z_q]$. If $i \notin \{p, q\}$ the term equals $\mathbb{E}[z_i^2]\mathbb{E}[z_p]\mathbb{E}[z_q] = 0$; if $i = p$ it equals $\mathbb{E}[z_p^3]\mathbb{E}[z_q] = 0$, and symmetrically for $i = q$. Hence $K_{pq} = 0$. On the diagonal, $K_{ii} = \mathbb{E}[v_i^4] + \sum_{j \neq i} \mathbb{E}[v_i^2 v_j^2] - (M+2) = \kappa_i + \sum_{j \neq i} \big(\mathbb{E}[v_i^2 v_j^2] - 1\big)$. □

Three consequences follow. First, the eigenvectors of $K(Z)$ still identify the structural directions under the common volatility factor that the co-kurtosis panel documents, even though that factor violates Assumption 1(ii); symmetry of z_t is not needed. Second, the eigenvalues equal the excess kurtoses only under exact independence; under common volatility each is shifted by the shock's excess co-kurtosis

with the others. The decomposition holds exactly in sample: with the co-kurtoses of Table F.4 it gives 4.797, 6.237 and 2.436 for v_1, v_2, v_3, against $d_i - \kappa_i = 4.793$, 6.235 and 2.435. Third, ordering by d_i and by the realised κ_i can differ in general; in these data they coincide. The all-ones weight of Kollo (2008) does not share the first property: its off-diagonal entries in the structural basis equal $2\big(\mathbb{E}[v_p^2 v_q^2] - 1\big) \neq 0$, so under common volatility its eigenvectors rotate away from the structural directions (Section D.2).

B.7 Reference rotation and posterior alignment

A reference rotation P_{ref} is computed once from a point estimate of the reduced form, and the sign convention is applied to it. Under the flat prior the posterior mean of $\{\Phi_j\}$ is the OLS estimate by construction, and at $T_{\text{eff}} = 9{,}058$ the posterior mean of Σ_u agrees with the OLS residual covariance to within 0.18 per cent on every element but the smallest off-diagonal entry, where the relative discrepancy is 0.71 per cent on an absolute magnitude of order 10^{-6}. The reference identification is therefore the estimation chain applied to the OLS residuals, an ICA-SVAR estimator in the sense of Lanne et al. (2017) and Gouriéroux et al. (2017), implemented through the FOBI eigendecomposition of Cardoso (1989), and the posterior draws produce the credible bands and not the point.

For each of the 5,000 posterior draws the per-draw rotation P_s is aligned to P_{ref} by greedy column matching on the absolute inner products $|P_s' P_{\text{ref}}|$, which fixes the permutation, followed by a sign flip whenever $\langle P_{s,k}, P_{\text{ref},k} \rangle < 0$. The first step addresses the column-swap problem that arises when two eigenvalues are close, so that a small perturbation reorders them; the second addresses the bimodality of each eigenvector's posterior induced by the sign indeterminacy. Both are label-fixing operations and both are applied to P and to the corresponding impact column in lockstep, so that $B^{(0)}\mathbf{v}_t$ is invariant. Without them, posterior medians of $B^{(0)}$ and of the impulse responses would mix opposite branches and distinct components, and would be meaningless.

B.8 The moving-block bootstrap at the daily stage

The posterior draws of Σ_u quantify uncertainty about the reduced-form covariance, but they treat the sample kurtosis matrix $\widehat{K}$, and hence the eigenvalues that order and identify the shocks, as if it were estimated from i.i.d. data. It is not. Daily financial innovations display strong volatility clustering, which leaves the rotation unbiased but inflates the sampling variance of $\widehat{K}$, and the posterior draws of Σ_u say nothing about that inflation. The moving-block bootstrap preserves serial dependence by construction, and it is the appropriate device precisely because it leaves point identification untouched while widening the bands to reflect the non-i.i.d. nature of the data. It is applied at two distinct points, with the same block length $\ell = 20$ trading days, chosen to span the typical persistence of daily volatility, roughly one trading month.

The first application is to the spectrum. The pre-whitened residual series is resampled in overlapping blocks to form 500 bootstrap samples, on each of which the full estimation chain is re-run. This yields bootstrap distributions of the eigenvalues and of the rotation columns, from which we report bootstrap confidence intervals for each λ_i; the bootstrap probability $\Pr^*(\lambda_1 > \lambda_2)$ of leading-shock dominance; and a column instability index, the average $1 - |\langle P_k^{(b)}, P_{\mathrm{ref},k} \rangle|$, measuring how stable each identified direction is across resamples. The last of these speaks most directly to identification strength, since a direction that survives resampling of the tail observations is identified by the structure of the data and not by a handful of days.

The second application is to the independence diagnostics. The same scheme is applied to the identified shock series themselves, with $B = 5{,}000$ replications, to obtain confidence intervals for the SV-robust co-moments. This step carries more weight than its routine appearance suggests. The odd-power statistics are the only evidence offered on contemporaneous independence, and their null values are zero, so everything turns on how far from zero a sample value has to be before it counts. Because the shocks inherit the volatility clustering of the data, an i.i.d. resampling scheme would produce intervals that are far too narrow and would manufacture rejections of an assumption that holds; blocking is what makes the diagnostic informative instead of mechanical.

Together the two applications separate the two things that can go wrong with an identification of this kind, an ill-determined spectrum and a failure of contemporaneous independence, and attach to each a measure of uncertainty that is valid under the dependence the data actually exhibit. Where they disagree with the posterior-Σ_u bands, the bootstrap is the quantity reported, since the posterior bands are constructed under an assumption the data reject.

B.9 The dimension of the daily system

The daily system is deliberately small. Because the identification operates at fourth order, dimension enters this design differently from the way it enters a conventional VAR, and two considerations fix the choice at $M = 3$.

The first is arithmetic. The identifying object is the fourth-cumulant tensor, with M^4 entries, 81 at $M = 3$ against 2,401 at $M = 7$, estimated from the same T. Standard intuition about VAR dimensionality concerns the M^2p conditional-mean coefficients and understates the problem here, because the parameter count of the identifying object grows twice as fast in the exponent, and the per-entry sampling error of a fourth moment is already large under heavy tails. The consequence falls on the distinct-kurtosis assumption and not on independence. Pairwise distinct kurtoses must be established for $M - 1$ adjacent pairs, two in three dimensions and six in seven, and a wider daily financial system supplies the additional components from blocks that are close to redundant: a set of yields and spreads is, in substance, a low-dimensional factor structure, so the extra components extracted from it are near-Gaussian, carry small and mutually similar kurtoses, and are separated from one another only weakly. Degeneracy at the foot of the spectrum, not dependence, is what a larger system produces.

The second consideration dissolves the apparent trade-off. A larger daily system would be motivated by the wish to capture the whole financial system and thereby avoid omitting channels. But in a two-stage design the daily VAR is an identification device whose only job is to span the directions that generate the tails; economic breadth is supplied by the ten-variable monthly macro system into which the shocks are injected. The term structure, the policy rate, credit quantities, housing and prices all enter there, at the frequency at which

they matter macroeconomically. Asking the daily block to do both jobs would degrade the one job it is uniquely able to do.

C The bridge to monthly frequency

C.1 No shock selection

Every shock identified in the daily stage travels to the macro stage, in its natural order $v_1, \ldots, v_M$. There is no selection step anywhere in the pipeline: the identification returns the M columns of the rotation in one go and all of them are carried forward.

This is a deliberate restriction on the analyst and not an omission. Selecting the shock to be instrumented on the basis of a statistic, its own kurtosis or its correlation with a real proxy, and then reporting that same statistic as evidence for the shock's importance, is circular. Keeping all shocks means the daily diagnostics remain independent measurements instead of selection criteria, and it is what allows the macro variance decomposition to report the importance of each shock without that importance having been built into which shock was carried forward. Economic meaning is assigned *ex post*, from the impulse-response signature, the sign anchor and the narrative record, and never upstream.

C.2 Temporal aggregation: the bridge filter

A daily shock cannot be used as a monthly instrument without an aggregation rule, and the rule is not innocuous. The bridge literature establishes that the high-frequency object must be aggregated with the filter matching how the low-frequency variable is sampled (Foroni and Marcellino, 2014; Marcellino, 1999): summing for flows, averaging for period-average series, skip-sampling for point-in-time stocks. The baseline uses the signed monthly mean,

$$Z_m^{(k)} = \frac{1}{D_m} \sum_{t \in m} v_{k,t}, \tag{C.1}$$

with D_m the number of trading days in month m.

The mean is the correct low-frequency filter for a linear VAR. Aggregating a linear daily VAR to monthly frequency makes the monthly reduced-form innovation, to a first order, an average of the daily innovations, so the monthly structural shock is proportional to the average of the daily ones, which is also the scheme of **?**. The sum

differs from the mean only by the factor D_m, which the normalisation on the reference variable removes up to its variation across months. The exact aggregation weights are not uniform: a shock early in the month affects more of the monthly average than one late in the month, which generates a triangular weighting (**?**), so (C.1) is the correct filter at first order and not exactly. This is the standard approximation in the bridge literature and we state it rather than conceal it.

The mean preserves what makes an object a shock. An instrument must inherit zero mean and approximate serial uncorrelatedness, and only the signed mean does. The absolute value and the square destroy the sign, are not zero-mean, are strongly serially correlated because volatility clusters, and, decisively, are not exogenous, since the intensity of one shock loads on the common volatility factor and hence on the other shocks. They are intensity measures suitable for the level-on-level validation and never instruments, and Table F.13 verifies this on all four candidate aggregates.

A property of the aggregate that must be declared. Since $\mathrm{Var}(Z_m) = \sigma^2/D_m$, an unbalanced number of trading days makes the instrument conditionally heteroskedastic by construction, independently of any heteroskedasticity inherited from the daily shocks. In this sample D_m ranges from 13 in January 1990, truncated by the $p = 9$ lags consumed in the daily VAR, to 23, a ratio of standard deviations of 1.33. This is one of the two reasons the inference of Section C.7 is a block bootstrap and not a Gaussian posterior band.

C.3 Sign re-anchoring at the bridge

The sign of each column of $B^{(0)}$ is a free normalisation. Replacing $(B^{(0)}, \mathbf{v}_t)$ by $(B^{(0)}S, S\mathbf{v}_t)$ with $S = \mathrm{diag}(s_1, \ldots, s_M)$, $s_j \in \{-1, +1\}$, leaves $B^{(0)}\mathbf{v}_t$ and hence the entire reduced form unchanged, because $S^2 = I$. Each s_j is therefore a separate and independent choice, and flipping one shock does not require flipping any other.

The daily stage anchors each shock so that its FEVD-dominant variable loads positively. That convention is the natural one for reading the impact matrix, but it has an inconvenient consequence for the macro stage: it makes v_1 and v_3 adverse, with credit stress up and volatility

up, while making v_2 expansionary, with equity up. Three impulse-response panels in which two shocks contract and one expands cannot be compared at a glance, and the reader has to mentally invert one of them.

The bridge therefore re-anchors the signs so that all three shocks are adverse, through a vector SIGN $= (+1, -1, +1)$ applied once, to the *daily* shock series, before any aggregation. The implementation matters in three respects.

1. Applying it to the daily series instead of to the monthly aggregates makes every downstream object consistent automatically. This is not cosmetic: the one-sided aggregates *swap* under a flip (POS $\mapsto$ $-$NEG), while the intensity measures abs and sq are even functions and are invariant. Applying the convention at the daily level makes both facts automatic.

2. The vector is stored with the aggregated instrument and read back by the macro stage instead of being declared twice, so the two stages cannot drift apart.

3. Because the daily results report the upstream convention, the sign of v_2 differs between the daily and the monthly parts of the paper: in the daily part v_2 raises equity, and from the macro section onward it lowers it. The one place where the bridge convention is used earlier is the signed panel of the external-validation table, whose caption says so, because a signed correlation against an uncertainty index is only interpretable once the shock is oriented adversely; the amplitude panel of the same table is invariant to the choice. Nothing else about the shock changes, and no result depends on which of the two conventions is adopted.

C.4 The monthly BVAR: prior and lag order

The monthly system is estimated in levels as

$$X_t = \sum_{j=1}^{p_m} A_j X_{t-j} + A_0 + \varepsilon_t. \tag{C.2}$$

Unlike the daily system, here a Minnesota prior is appropriate: the sample is two orders of magnitude shorter and the identification does

not rely on non-Gaussianity, so the internal-consistency objection of Section B.1 does not apply.

Mixed integration orders are handled through the prior mean. Following Litterman (1986), Giannone et al. (2015) and Sims and Zha (1998), the prior mean on the first own lag is set per variable,

$$\delta_i = \begin{cases} 1 & \text{random-walk prior mean, } I(1) \text{ variables,} \\ 0 & \text{white-noise prior mean, } I(0) \text{ variables.} \end{cases} \tag{C.3}$$

The three financial series receive $\delta_i = 0$ and the remaining seven $\delta_i = 1$; in the augmented system the instrument itself receives $\delta = 0$, which is the correct prior for an object that is supposed to be white noise. Estimating a mixed $I(0)/I(1)$ system in levels is consistent by Sims et al. (1990), and setting the prior mean by integration order is the coherent Bayesian treatment of the mixture, which avoids both the information loss of differencing and the misspecification of imposing a random walk on a mean-reverting risk premium.

The lag order is fixed a priori, not selected. $p_m = 15$ throughout. This is a reallocation of what the data decide and not a retreat from letting them decide.

Under a shrinkage prior, lag length ceases to be a model-selection problem. The Minnesota prior variance decays as $1/s^2$ in the lag s, so coefficients on distant lags are shrunk toward zero by the data, through hyperparameters that are themselves chosen by maximising the marginal likelihood. The choice therefore moves from a discrete and badly-identified dimension, *which* p, along a likelihood surface that Giannone et al. (2015) show to be nearly flat once the shrinkage is optimised, to a continuous and well-identified one, *how much* shrinkage. In this application the optimised hyperparameters are $\tau^* = 3.36$, $\lambda^* = 4.04$ and $\mu^* = 6.30$, none of them imposed.

The asymmetry of the two errors then settles the level. A generous p imposes a soft restriction: the coefficient on lag 14 is pulled toward zero but the data may override it. A short p imposes a hard and unrevisable one: the coefficient on lag 10 is exactly zero with infinite confidence. Bańbura et al. (2010) use thirteen lags for monthly data on this reasoning, and fifteen adds a further margin. The claim is verified in Section F.8.

C.5 Internal against external, and against local projections

Let Z_t denote the aggregated shock for one of the M daily shocks. The macro stage estimates the augmented system

$$\widetilde{X}_t = \left(Z_t,\ X_t'\right)' \in \mathbb{R}^{N+1}, \qquad \widetilde{X}_t = \sum_{j=1}^{p_m} \widetilde{A}_j \widetilde{X}_{t-j} + \widetilde{A}_0 + \tilde{\varepsilon}_t, \tag{C.4}$$

with the instrument ordered first, and identifies recursively,

$$\widetilde{\Sigma} = PP', \quad P \text{ lower triangular}, \qquad b \ = \ \frac{P_{\cdot,1}}{P_{1,1}}. \tag{C.5}$$

Because Z_t is ordered first and is contemporaneously exogenous, the first Cholesky shock of the augmented system is the shock the instrument proxies, and the normalisation by $P_{1,1}$ expresses the response per unit movement of Z. This is the internal-instrument identification of Plagborg-Møller and Wolf (2021). Each of the three shocks is instrumented in its own augmented system, so no recursive ordering is imposed among the instruments: their mutual orthogonality is inherited from the daily identification.

Why internal rather than external. The external proxy-SVAR identifies the impact vector by projection, $b_{\text{ext}} = \mathbb{E}[\varepsilon_t Z_t]/\mathbb{E}[Z_t^2]$ (Mertens and Ravn, 2013), and is consistent under relevance and exogeneity provided the shock is invertible from the macro VAR's residuals. Plagborg-Møller and Wolf (2021) prove that under invertibility the two estimators recover the same normalised impulse responses, and that without invertibility the internal estimator still identifies them while the external one does not; measurement error in the instrument does not by itself separate the two. The internal design therefore weakly dominates: it agrees with the external estimator wherever the external one is valid, and survives where it is not. It also identifies from the innovation of Z_t, so the partial predictability of the credit instrument documented in Section C.9, which would contaminate an external use of the instrument, is filtered out before the impact vector is formed.

Why not local projections. Plagborg-Møller and Wolf (2021) also show that the internal-instrument estimand is the LP-IV estimand of Stock and Watson (2018), so the choice between them is one of

estimation and not of identification. At $h = 0$ the two estimators coincide exactly when the local projection controls for the VAR's own lags; at $h > 0$ they differ only in that the VAR ties the horizons together through its estimated dynamics whereas the local projection estimates each horizon by a separate regression (Jordà, 2005). Because the two share an estimand, the choice is one of estimation: the local-projection counterpart of the internal-instrument VAR controls for lags of both the instrument and the system and is equally robust to non-invertibility. The VAR is preferred for precision, since up to horizon p_m it pools information across horizons, at the cost of the extrapolation bias beyond p_m documented by Li et al. (2024). The VAR structure also does useful work at $h \geq 1$: the impact vector b enters every horizon through $\Psi_h b$, so the well-identified financial entries of b carry the poorly identified real ones, and only the impact month depends on a single element of b in isolation.

What is actually identified. This is the interpretive consequence of the design and it must be stated explicitly, because it is easy to overclaim. The first equation of (C.4) is

$$Z_t \;=\; c + \sum_{j=1}^{15} a_j' \widetilde{X}_{t-j} \;+\; \eta_t, \tag{C.6}$$

so the object entering (C.5) through $\widetilde{\Sigma}$ is η_t: the component of the monthly aggregate that could not have been predicted from fifteen months of the entire system's history. The identified shock is the unanticipated component of the monthly aggregate and not the aggregate itself. The consequences run in two directions. Any predictable component of Z_t, such as the serial correlation documented in Table F.13, is filtered out by the estimation instead of being carried into the identification, which is a further reason the internal design is the robust one here. And, less comfortably, the effective instrument is *smaller* than Z_t: with part of the aggregate predictable, the innovation has less variance than the raw series, which is visible in the first-stage statistics.

C.6 Normalisation of the responses

Equation (C.5) returns responses per unit movement of Z, and Z has no natural unit: its standard deviation is about 0.21, so one unit is a

shock of roughly five standard deviations and the implied responses read as implausibly large. Every reported response is therefore rescaled. Let ref index a reference variable and b_{ref} its impact response under (C.5); responses are multiplied by

$$c \;=\; \frac{\gamma}{\left|b_{\text{ref}}\right|}, \qquad \gamma = 0.25, \quad \text{ref} = \text{EBP}, \tag{C.7}$$

so that every shock is expressed as the shock that moves the excess bond premium by 25 basis points on impact.

This is the only scaling the estimator identifies. The point is easy to miss because (C.7) looks like a choice of units. Plagborg-Møller and Wolf (2021) show that when the instrument is contaminated by measurement error unrelated to the shock of interest, which is the case that motivated the internal design in the first place, the internal-instrument estimator attenuates every response by a common factor, and that the attenuation is identical at all horizons and for all response variables. Writing $Z_t = \phi\,\varepsilon_{1,t} + \nu_t$ with ν orthogonal to the shock, the Cholesky-first vector recovers

$$b_i \;=\; \Theta_i\,\lambda, \qquad \lambda \;=\; \frac{\phi\,\sigma_\varepsilon^2}{\phi^2\sigma_\varepsilon^2 + \sigma_\nu^2}, \tag{C.8}$$

where λ is the signal-to-noise ratio of the instrument. What follows from (C.8) settles the choice.

1. **Scalings expressed in the instrument's own units are not comparable across shocks.** Reporting b_i itself, per unit of Z, or $b_i\,\text{sd}(Z)$, per one standard deviation, leaves λ in the reported number. Since λ differs across the three instruments, whose first-stage F statistics on the reference variable range from 11.5 to 21.7, a comparison on that scale would systematically shrink whichever shock has the noisier instrument. Under the one-standard-deviation convention the effect is explicit: since $\text{sd}(Z)^2 = \phi^2\sigma_\varepsilon^2 + \sigma_\nu^2$, the reported response is $\Theta_i\,\phi\sigma_\varepsilon^2/\text{sd}(Z)$, which falls as the instrument becomes noisier, which is the opposite of what a comparison across shocks should do.

2. **A ratio to a variable inside the system cancels it.** The reported object under (C.7) is $\gamma\,\Theta_i\lambda/|\Theta_{\text{ref}}\lambda| = \gamma\,\Theta_i/|\Theta_{\text{ref}}|$: the attenuation appears in numerator and denominator and disappears.

This is the relative impulse response, and it is exactly the object Plagborg-Møller and Wolf (2021) prove the internal instrument identifies. Stock and Watson (2018) reach the same estimand from the local-projection side under the name of the unit effect normalisation.

The precedent is direct. Plagborg-Møller and Wolf (2021) report the response of the excess bond premium to a monetary shock normalised to move the one-year rate by 100 basis points on impact, in the same internal-instrument framework, and Gertler and Karadi (2015) use the same device when comparing specifications, though their baseline figures report a one-standard-deviation shock, which happens to move that rate by about 25 basis points.

Choosing the reference variable. Because the denominator of (C.7) is estimated, the choice of ref is not innocuous, and Gertler and Karadi (2015) flag the failure mode themselves: normalising on a variable that a given shock barely moves inflates that shock's responses. The reference is therefore chosen by a stated criterion. A candidate must be moved in the same direction by all three shocks, since a reference that one shock lowers cannot put the three panels on a common axis, and it must be moved by a comparable amount, since a large spread in $|b_{\text{ref}}|$ is what produces the inflation. Four of the ten macro variables satisfy the first condition; on the second the excess bond premium is the best of them, its impact varying by a factor of 1.98 across the three shocks against 3.09 for the risk-aversion index and 5.22 for the uncertainty index. The funds rate has a smaller spread still but fails the first condition, its sign differing across shocks. EBP is also in percentage points and is the standard credit-risk price in the literature (Gilchrist and Zakrajšek, 2012), but the criterion selects it before those considerations are brought to bear.

Two further features of (C.7) are deliberate. The denominator is an absolute value, because normalising to $b_{\text{ref}} = +\gamma$ would force the reference variable to rise for every shock and would therefore overwrite the sign convention for any shock that lowers it; the scale factor is positive by construction and the direction of the response is left entirely to the sign anchor. And the reference is common to all shocks and is not chosen by first-stage strength. A per-shock reference would put the three panels on three different yardsticks, literally answering three different questions, and choosing it by the largest first-stage F would

be worse, since F measures precision and not magnitude: a shock whose impact on its reference is small but precisely estimated would receive a large scale factor. Applied to these data that rule spreads the three scale factors over a 5.5-fold range against the 2.0-fold range of the common reference.

The normalisation does not make magnitudes comparable across shocks in any deeper sense than the units it imposes, which is why the ranking exercise in the paper rests on the variance decomposition and not on peak responses. What (C.7) delivers is that the object being compared is identified at all.

C.7 Inference: moving-block bootstrap on the shrunk estimator

The Gaussian–inverse-Wishart posterior of the augmented BVAR prices no conditional heteroskedasticity. The instruments are conditionally heteroskedastic for two independent reasons, the calendar effect $\mathrm{Var}(Z_m) = \sigma^2/D_m$ of Section C.2 and the volatility clustering inherited from the daily shocks, so a band that assumes homoskedastic Gaussian innovations is not the right object, particularly at impact, where the identification does its work.

Inference is therefore a residual-based moving-block bootstrap of the augmented VAR, valid under conditional heteroskedasticity and heavy tails (Brüggemann et al., 2016; Jentsch and Lunsford, 2019), with block length $\ell = \mathrm{round}(5.03\,T^{1/4}) = 23$ months and $B = 2{,}000$ replications, blocks recentred. No replication was lost for any of the three shocks.

The estimator replicated must be the estimator reported. A bootstrap band describes the sampling variability of whatever is re-estimated inside it. Re-estimating by OLS while reporting a shrunk Minnesota point estimate would pair a line from one estimator with an interval from another, and at $p_m = 15$ the unshrunk estimator is not merely different but unusable, with 166 coefficients per equation and 414 observations. The obstacle is only apparent. With a conjugate Normal–inverse-Wishart prior written as dummy observations (Y_d, X_d), the posterior mean of the coefficients is ordinary least squares on the

augmented data,

$$\widehat{A}_{\text{minn}} = \Big(\big[X; X_d\big]' \big[X; X_d\big] \Big)^{-1} \big[X; X_d\big]' \big[Y; Y_d\big], \tag{C.9}$$

so re-estimating the shrunk estimator inside a bootstrap replication costs one slightly larger linear solve. Every replication is therefore shrunk, and point and band belong to the same estimator while the band remains heteroskedasticity-robust. The identity is verified rather than assumed: the full-sample check compares (C.9) against the simulated posterior mean in units of posterior standard deviation, since the benchmark is itself a Monte Carlo average and a machine-precision tolerance would reject a correct replication. The realised maximum deviation is 0.10 posterior standard deviations, about 3.2 Monte Carlo standard errors, which is the order of magnitude expected for a maximum over some 1,800 coefficients.

The normalisation is applied inside the bootstrap. This is the one point at which the reporting convention interacts with inference, and getting it wrong is invisible. Because the reported response is the ratio $\gamma\,\Theta_i/|\Theta_{\text{ref}}|$, its sampling distribution reflects uncertainty about the denominator as well as the numerator. Computing a band under one scaling and then multiplying it by a factor read off the point estimate produces an interval for a different object: the unit effect normalisation of Stock and Watson (2018) is a ratio, and its sampling uncertainty must be priced as such. Each replication is therefore rescaled by its own impact on the reference variable before the percentiles are taken, and the same is done draw by draw for the Bayesian posterior.

The correction has two visible effects. First, the spread of the scale factor across replications is itself a diagnostic, and it is the quantity an outside rescaling discards. Its $[16, 84]$ range is [0.49, 0.79] for the credit shock, [0.82, 2.09] for the equity shock and [0.60, 0.97] for the volatility shock, against point values of 0.59, 1.16 and 0.75. The equity shock has the least well determined denominator, which is the arithmetic counterpart of its having the smallest impact on EBP. In no case does the sign of the reference response flip in more than 0.2 per cent of replications, so the ratio is well behaved and the Cauchy-tail pathology that afflicts weak-instrument ratios does not arise here. Second, the bands change. Relative to the incorrect outside rescaling they are wider by a median factor of 1.08 for the credit shock, 1.29 for the equity shock

and 1.03 for the volatility shock, the ordering tracking the dispersion of the scale factor exactly, and at short horizons for the credit shock they are slightly *narrower*, because numerator and denominator are positively correlated across replications and the ratio varies less than the numerator alone. A constant multiplier could produce neither effect. No conclusion is altered: across all 1,440 combinations of shock, variable and horizon, not one changes whether its 68% interval excludes zero.

The Gaussian band is retained for comparison. It is overlaid on every figure. The robust band is the wider of the two at impact, by a median factor of 1.16, 1.12 and 0.92 across variables for the three shocks and by 1.8 for industrial production under the credit shock, and the narrower at long horizons, by factors of 0.88, 0.94 and 0.77. The two are not nested: the bootstrap conditions on the estimated innovation covariance and varies through resampled residuals, while the posterior integrates over parameter and covariance uncertainty. Both are reported, and neither is presented as a correction of the other.

A limitation, stated plainly. The bands price uncertainty about the monthly VAR parameters and about the residual process. They do not propagate uncertainty about the daily identification itself: the instrument is treated as the point-estimated shock series and not as a distribution over the posterior draws of the daily rotation. The reported intervals are therefore conditional on the daily identification, and understate total uncertainty by an amount this design does not measure. The daily rotation is stable under its own block bootstrap, with column instability between 0.040 and 0.077, which bounds the likely magnitude of the omission without eliminating it. Propagating it through matched draws, one draw of the daily rotation paired with one draw of the monthly BVAR, is feasible at essentially no additional cost.

C.8 Instrument relevance and the weak-instrument question

Relevance is assessed by regressing each macro variable's reduced-form residual on the demeaned instrument and computing the heteroskedasticity-robust effective F of Montiel Olea and Pflueger

(2013), which for a single instrument and a single endogenous variable coincides with the robust squared t-statistic. Two reporting conventions matter. The statistic is reported for the variable used in the normalisation, named explicitly, and not as a maximum over the ten macro variables, because a maximum over ten tests is a selected maximum and is biased upward. And the F against a slow real variable is not the relevant object, for the structural reason set out in the paper. Both statistics are reported, with their interpretations separated.

Where the weak-instrument concern does and does not bite. The internal-instrument design changes the answer relative to the external proxy-SVAR for which the standard weak-instrument apparatus was built. The response per unit of the instrument innovation is not a weak-instrument ratio at all: by (C.5) its denominator is $P_{1,1}$, the standard deviation of the instrument's own innovation, which is bounded away from zero by construction. The familiar pathology, an estimator that is a ratio whose denominator may be near zero, producing Cauchy-like tails and a Wald interval that under-covers invisibly, is created here entirely by the normalisation, whose denominator is the impact on the reference variable. That quantity is estimated with an effective F between 11.5 and 21.7, or a robust t between 3.4 and 4.7: above the conventional threshold, though not so far above it that the denominator can be treated as known. The design responds directly rather than through a separate exhibit. The normalisation is applied inside every bootstrap replication, so the reported percentile band is a band on the ratio and prices the denominator's sampling variability, and the distribution of the scale factor across replications is the empirical answer to whether that denominator is well determined.

On weak-instrument-robust sets. The analytical Anderson–Rubin construction of Montiel Olea et al. (2021) is the off-the-shelf device for this problem, and it is not used as the reported interval, for a reason of scope and not of principle: it is derived for the *external* proxy-SVAR, with macro-only dynamics and the projection estimator, so a set constructed that way bounds a different estimator from the one reported here. A confidence set for one estimator is not a confidence set for another, and presenting it alongside the internal-instrument line would invite exactly that conflation.

C.9 Shock predictability: the test

A structural shock must be unpredictable from information available at $t-1$. If part of Z_t was already forecastable, that part is not a surprise: either the aggregate is contaminated by anticipated components, or the daily VAR from which it was extracted omits information that the macro system contains. The test below is a necessary check on the first possibility; it is not a test of informational sufficiency in the sense of Forni and Gambetti (2014).

Testing against the daily VAR's own variables would be vacuous, since the shock is orthogonal to their lags by construction of least squares, so the test uses information the daily system does not contain. Each instrument is regressed on two lags of four macroeconomic variables,

$$Z_t = c + \sum_{l=1}^{2}\Big[\beta_{1l}\Delta \ln \mathrm{INDPRO}_{t-l} + \beta_{2l}\Delta \mathrm{UNRATE}_{t-l} + \beta_{3l}\Delta \ln \mathrm{PCEPI}_{t-l} + \beta_{4l}\,\mathrm{EBP}_{t-l}\Big] + e_t, \quad \text{(C.10)}$$

and joint significance is tested. This is the standard validity check applied to identified and narrative shock series (Ramey, 2016), and the analogue of the predictability checks run on high-frequency surprises by Gertler and Karadi (2015).

Two aspects of the implementation are substantive. The $I(1)$ macro variables enter in first differences and the $I(0)$ financial variable in levels, because regressing a stationary instrument on near-integrated regressors would give a non-standard F distribution and an R^2 inflated by shared persistence; this concerns the test only, and the VAR itself is estimated in levels with the unit-root prior of (C.3). And inference is heteroskedasticity-robust, using an HC1 Wald statistic, because the homoskedastic F over-rejects on an instrument known to be conditionally heteroskedastic.

Reporting individual t-statistics alongside the joint test is what makes the exercise informative instead of a gate. If the test rejects, the individual coefficients say which information predicts the shock, and the reading differs sharply: predictability from lagged EBP indicates persistence of credit conditions themselves, which is benign, whereas predictability from lagged real activity indicates that the financial shock is partly endogenous to the business cycle, which constrains the label.

What a rejection does and does not imply. Two limits are worth fixing. The test is a *necessary* condition, so passing it does not establish exogeneity with respect to the other structural shocks. And it is a test on the *level* of Z_t, whereas by (C.6) the identification uses the innovation η_t, which is orthogonal to fifteen lags of the whole system by construction; a rejection is therefore stringent for an external use of the instrument and only indirectly informative about the internal design adopted here. What a rejection constrains is the economic label attached to the shock. The full test output is in Table F.14.

C.10 Forecast error variance decomposition

An impulse response states how a variable moves after a shock; it does not state whether the shock matters. A response of a given size from a disturbance accounting for two per cent of a variable's forecast error variance is a true but economically marginal statement; the same response from a shock accounting for a quarter of it is a first-order result. The two are separated by the variance decomposition, which is also invariant to the normalisation of Section C.6.

The augmented system (C.4) with a complete Cholesky factorisation delivers $N+1$ orthogonal shocks whose contributions to the h-step forecast error variance sum exactly to one, the first being the identified shock. Writing Ψ_s for the moving-average matrices of (C.4) and $\Theta_s = \Psi_s P$,

$$\mathrm{FEVD}_n(h) \;=\; \frac{\sum_{s=0}^{h} \Theta_s(n,1)^2}{\sum_{s=0}^{h} \sum_j \Theta_s(n,j)^2}. \tag{C.11}$$

Only the first share is reported, because the remaining N orthogonal shocks carry no economic label. The three points below are recorded because an error in any of them would be silent.

The scaling in (C.11) is not the scaling of the impulse response. The response uses $b = P_{\cdot,1}/P_{1,1}$, a reporting convention; the decomposition uses $P_{\cdot,1}$ itself, the impact of the shock with unit variance, which is the only scaling under which the squared contributions of the orthogonal shocks sum to the total forecast error variance. Using the response vector in the numerator and the true mean squared error in the denominator would inflate every share by $1/P_{1,1}^2$ with no visible symptom. Since $\widetilde{\Sigma} = PP'$, the identity $\sum_j \Theta_s(\cdot,j)^2 = \mathrm{diag}(\Psi_s \widetilde{\Sigma} \Psi_s')$

delivers numerator and denominator from a single matrix product.

An exact internal check. The first row of the decomposition is the instrument itself. Because P is lower triangular, at $h = 0$ the first Cholesky shock explains Z entirely, so that entry must equal one exactly. It is verified at every run rather than assumed, and any deviation would indicate an error in the recursion or the ordering. Shares are also not additive across shocks, since each shock is instrumented in its own augmented system and the three decompositions come from three different Cholesky factorisations.

The reported shares are lower bounds. This follows from the same attenuation that motivated (C.7), and it is the one respect in which the decomposition is not fully identified. The first Cholesky shock of the augmented system is the innovation to Z, which by (C.8) contains the measurement error as well as the structural shock; normalised to unit variance its impact vector is ρ times the true one, where $\rho = \mathrm{Corr}(u_Z, \varepsilon_1) \leq 1$. The numerator of (C.11) is therefore scaled by ρ^2 and the denominator is not, so $\widehat{\mathrm{FEVD}} = \rho^2 \,\mathrm{FEVD}$. Plagborg-Møller and Wolf (2022) show that in this situation the variance decomposition is interval-identified rather than point-identified, with the proxy delivering the lower end of the interval. The reported shares understate the true contribution of each shock, so a large share is a conservative finding and not an optimistic one. But ρ differs across the three instruments, so the decomposition is not perfectly comparable across shocks either, less badly than a magnitude comparison since it is at least free of the reporting normalisation, but the caveat is real.

C.11 Computational implementation

Data preparation, comprising the download from FRED and LSEG, the merge, the NYSE calendar alignment, the construction of the Baa−Aaa spread and the assembly of the monthly macro panel, is carried out in Python 3 using `pandas`, `numpy`, `fredapi` and `pandas_market_calendars`. The BVAR estimation, posterior simulation, Mórі identification, near-degeneracy test, moving-block bootstrap and the bridge to monthly frequency are implemented in MATLAB using the `bvar_` toolbox of Ferroni and Canova (2021).

At the daily stage, posterior inference uses 5,000 draws of Σ_u, with no draw discarded for numerical ill-conditioning. The impulse-response horizon is $H = 20$ trading days. The block bootstrap uses $B = 500$ replications with block length $\ell = 20$ for the spectrum, and $B = 5{,}000$ replications at the same block length for the independence diagnostics.

At the monthly stage, the augmented BVAR is estimated on 429 months, 414 usable observations at $p_m = 15$, giving 166 coefficients per equation in the augmented system, with 1,000 posterior draws per shock and a horizon of 48 months. The moving-block bootstrap uses $B = 2{,}000$ replications with block length $\ell = 23$ months, all of them usable for each of the three shocks. The Minnesota hyperparameters are optimised once on the macro-only system and reused for the three augmented systems, so that the three shocks are not implicitly given three different degrees of shrinkage.

D The choice of contraction: Móri versus Kollo

The identification contracts the fourth-cumulant tensor with the identity weight, following Móri et al. (1993). The estimator most often cited in this literature is instead Kollo (2008), which uses the all-ones weight. This section sets out the formal relationship between the two and documents why the choice is not innocuous in this sample.

D.1 The star product and the choice of weight

The star product is the contraction of two of the four indices of the cumulant tensor against a symmetric weight A, and Lemma 1 of the paper establishes that for any such weight

$$\widehat{M} \;=\; W\,\mathrm{diag}\big(\kappa_r\, c_r\big)\,W', \qquad c_r \;=\; w_r' A\, w_r, \tag{D.1}$$

where w_r is the r-th column of W. Since W is orthogonal this is already a spectral decomposition, so the eigenvectors are the columns of W whatever weight is used: contracting the tensor costs nothing in the identifying direction, because under Assumption 1 the tensor is diagonal in the structural basis and any contraction returns $W\mathrm{diag}(\cdot)W'$. The only requirement is that the diagonal weights $\kappa_r c_r$ be distinct, so that the eigendecomposition is unique. The eigenvalues depend on the weight, and only through c_r.

Table D.1: The two contractions of the fourth-cumulant tensor. $s_r = \mathbf{1}'w_r$ is the sum of the r-th column of W, that is, the projection of shock r onto the all-ones direction of the pre-whitened space.

Estimator	Weight A	$c_r = w_r' A w_r$	Eigenvalues of $\widehat{M}$
Móri $K(Z)$	I	$\lVert w_r\rVert^2 = 1$	κ_r
Kollo $C(Z)$	$\mathbf{1}\mathbf{1}'$	$(\mathbf{1}'w_r)^2 = s_r^2$	$\kappa_r\, s_r^2$

For Móri's weight, $c_r = 1$ exactly, because the columns of the orthogonal matrix W are unit vectors, so the eigenvalues of $K(Z)$ *are* the excess kurtoses of the structural shocks. For Kollo's all-ones weight the summed indices i and j are free and each contributes $\sum_i W_{ir} = s_r$, so the eigenvalues are the kurtoses rescaled by s_r^2.

Rotation-equivariance settles the choice. Consider a change of pre-whitening basis $\boldsymbol{\iota} \mapsto Q\boldsymbol{\iota}$ for orthogonal Q, for instance replacing the Cholesky factor by the symmetric square root, or reordering the variables. The identity weight is invariant under conjugation, $Q'IQ = I$, so $K(Z) \mapsto QK(Z)Q'$ and its eigenvalues are unchanged. This is as it must be, since the excess kurtosis of shock r is a property of the shock and not of the arbitrary basis used to whiten the residuals. The invariance is not merely formal in this sample: recomputing the whole reference identification with the symmetric square root in place of the Cholesky factor leaves the eigenvalues d_i, the realised kurtoses κ_i and the impact matrix $B^{(0)}$ unchanged to four decimal places. The all-ones weight is not invariant, since $Q'\mathbf{1}\mathbf{1}'Q \neq \mathbf{1}\mathbf{1}'$, so both the magnitudes and the ordering of Kollo's eigenvalues depend on the whitening basis and ultimately on the arbitrary ordering of the variables in the Cholesky factorisation. Móri's contraction is the coordinate-free choice.

D.2 Is Móri too strict?

It is sometimes suggested that Kollo's $C(Z)$ is preferred in applied work because it affords greater generality. Equation (D.1) shows this to be backwards, on three counts.

The two estimators impose *identical* assumptions on the data generating process: mutual independence, at most one Gaussian component, and distinct diagonal weights. Kollo does not purchase any degree of freedom; it changes a weight.

In population, with exact independence and distinct kurtoses, the two recover the *same shocks*, because they share eigenvectors. Any empirical divergence between them is therefore a symptom of finite-sample noise in $\widehat{K}$ or of imperfect independence, and not a difference in what they target.

The equivalence holds under exact independence. Under a common volatility factor, which the co-kurtosis panel documents, the Kollo matrix acquires off-diagonal terms $2\big(\mathbb{E}[v_p^2 v_q^2] - 1\big)$ in the structural basis and its eigenvectors rotate away from the structural directions, whereas the Móri matrix stays diagonal (Lemma B.1). The identity weight is therefore not only basis-invariant but also robust to the one dependence the data exhibit.

Kollo's genuine advantage is narrow and double-edged. The factor s_r^2 can break a tie when two shocks have nearly equal kurtoses, a case in which Móri's contraction is degenerate, because $s_i \neq s_j$ may separate them. By the same mechanism it can reorder spuriously, on the basis of a quantity that is a property of the whitening basis and not of the shocks. This is a different configuration of degenerate cases, of measure zero either way, and not more freedom. In the present application the estimated kurtoses are well separated, so the degeneracy Kollo was designed to repair does not arise, and the contaminant it introduces does.

D.3 What s_r measures, and why it is not the systemic factor

The factor $s_r = \mathbf{1}'w_r$ has a concrete reading: it is the projection of shock r onto the direction $(1, 1, \ldots, 1)$ of the pre-whitened space, that is, how strongly the shock moves all variables in the same direction. Kollo's weighting therefore up-weights co-moving directions, which invites the interpretation that $C(Z)$ ranks first the systemic factor and might be preferable when the object of interest is a common factor.

The interpretation does not survive contact with the economics. A financial crisis moves variables with *mixed* signs, equity down, volatility up, credit spreads up, a direction $d = (-1, +1, +1)$ and not $(1, 1, 1)$. The naive sum $\mathbf{1}'w_r$ cancels the equity decline against the credit and volatility increases instead of accumulating them. Empirically, in the baseline identification, the shock with the largest s_r is the *equity* shock ($s_2 = 1.66$), whose signature is risk-on, while the credit shock that carries the crisis signature has $s_1 = 0.49$ and the volatility shock has $s_3 = 0.11$. Kollo's weight promotes the anti-crisis direction. Moreover s_r lives on P, in the pre-whitened space, and not on the impact matrix $B^{(0)}$ in economic units, so it is doubly removed from any economic direction. A genuine systemic-factor objective would require an explicitly targeted approach, projecting on d or using a max-share criterion, and not the sum $\mathbf{1}'w$.

D.4 Placement in the ICA literature

An honest framing of the literature is that a single contraction is a computational shortcut and not a gold standard. The FOBI algorithm

of Cardoso (1989) uses precisely a Móri-type contraction, a single diagonal contraction with kurtosis weights, so $K(Z)$ *is* the classical FOBI object. The JADE algorithm of Cardoso and Souloumiac (1993), the robust standard, does not use a single contraction at all: it jointly diagonalises the full set of cumulant matrices, exploiting the entire tensor instead of one scalar summary. The $C(Z)$ matrix of Kollo (2008) is a specific single-matrix proposal with the all-ones weight, introduced to repair the degeneracy of $K(Z)$ under identically distributed components, a case that does not arise here. If the concern is robustness, the principled answer is joint diagonalisation and not a different single weight. The present analysis uses $K(Z)$ as the baseline, reports $C(Z)$ in parallel as a robustness check, and flags JADE-style joint diagonalisation as the natural extension.

D.5 The empirical comparison

The paper reports the two contractions evaluated at the reference identification. That table supports three readings, set out here in full.

(1) The Móri eigenvalues and the kurtoses. The eigenvalues d_i and the realised kurtoses κ_i coincide under exact independence, and in the data they agree exactly in ordering, which is what the identification consumes. The levels differ, and Lemma B.1 gives the gap exactly: $d_i - \kappa_i = \sum_{j \neq i}(\mathbb{E}[v_i^2 v_j^2] - 1)$, which the estimates reproduce to the third decimal (4.79, 6.24 and 2.44 for v_1, v_2, v_3). The gap is therefore the excess co-kurtosis, a measure of the common volatility factor, and not a diagnostic on contemporaneous independence. Regularisation plays no role, because none is applied ($\tau = 0$ in (B.5)); and because K_z is invariant under orthogonal conjugation, a ridge would divide all three d_i by the same factor and leave the comparison intact.

(2) The Kollo weights reverse the ranking. The credit shock has the largest excess kurtosis in the system by a factor of three, 37.9 against 12.4, yet its projection on the all-ones direction is small, $s_1 = 0.49$, so $C(Z)$ assigns it 9.22 and demotes it to second place. The equity shock, with less than a third of the credit shock's kurtosis, has $s_2 = 1.66$ and is promoted to first with 34.04. Ordering by the Kollo eigenvalue would therefore lead one to call the equity shock the leading, most non-Gaussian shock when it is nothing of the sort. The reversal

is not marginal: the ratio $\kappa_1/\kappa_2 = 3.06$ becomes $34.04/9.22 = 3.69$ in the opposite direction.

(3) The economics confirms which ranking is right. One could object that a reordering is merely a relabelling. It is not, because every downstream object, the economic reading, the shocks carried to the macro stage, the narrative interpretation, depends on which shock is called leading. And the data adjudicate. The credit shock is the one whose realisations spike at Lehman and COVID, the one that carries the canonical crisis signature, and the most non-Gaussian component of the system. The shock Kollo would promote is the equity shock, whose positive direction is risk-on. The basis-dependent factor s_r^2 does not encode a systemic-factor interpretation; it encodes alignment with the direction $(1, 1, 1)$, which in a system containing a spread and a volatility index is not a crisis direction at all.

E Existence of the fourth moment

The paper reports Hill indices below 4 and notes that this implies a potentially infinite fourth moment. That implication presupposes a genuine Pareto tail, and it fails for distributions that are heavy-tailed in finite samples while possessing moments of every order, in particular for normal mixtures with rare large jumps, for which the Hill estimator returns a finite index well below 4.

The two hypotheses are separable in the data, because they make different *joint* predictions about the tail index and the sample kurtosis. Fitting a Student-t to match the observed Hill index alone, and then reading off the sample excess kurtosis it implies, gives Table E.1.

Table E.1: Discriminating a power law from a finite-moment mixture. The mixture is $v = (z + \mathbf{1}\{U < p\}\eta)/\text{sd}$ with $z \sim N(0,1)$ and $\eta \sim N(0, \sigma_\eta^2)$, calibrated by grid search on (p, σ_η) to match *both* moments. The Student-t column matches the Hill index only.

	Observed		Fitted jump mixture				Student-t
Component	$\hat{\alpha}$	Kurt.Exc	p	σ_η	$\hat{\alpha}$	Kurt.Exc	Kurt.Exc
SP500	2.65	9.99	0.032	4.00	2.75	10.37	29.5
VIX	2.89	9.26	0.028	4.00	2.94	9.86	16.4
CSPREAD	2.38	34.35	0.020	7.00	2.33	35.29	61.0

A Student-t whose Hill index matches the observed one, requiring $\nu = 2.99,\ 3.37,\ 2.59$ respectively, implies sample excess kurtoses between 1.8 and 2.9 times those actually observed, and overshoots in the same direction for all three components. The jump mixture matches both statistics to within a few per cent, and its *population* excess kurtoses, 10.32, 9.96 and 36.03, are finite.

The evidence therefore favours the finite-moment regime, in which the fourth-cumulant estimator is consistent, and the Hill values are best read as a measure of tail thickness and not as a verdict on moment existence. Two qualifications preserve the main text's use of them. Heavy tails inflate the sampling variance of $\widehat{K}$ whether or not the moments exist, and the baseline meets that with the block bootstrap and not with the ridge of Section B.4, which is set to $\tau = 0$ precisely because a deterministic rescaling of the point estimate does not address a variance problem. And because the regime cannot be established with certainty, the inferential device is chosen to be valid in both.

F Additional results and diagnostics

All objects in this section are computed on the baseline three-variable system at $p = 9$ with $T_{\text{eff}} = 9{,}058$, unless stated otherwise.

F.1 Lag selection at the daily stage

Table F.1: Flat-prior log-marginal likelihood for candidate lag orders, baseline system. $T_{\text{eff}} = T - p$; $k = Mp + 1$ parameters per equation; $\Delta = \ln \text{ML}(p) - \max_{p'} \ln \text{ML}(p')$. The specification $p = 9$ is selected.

p	T_{eff}	k	$\ln \text{ML}$	Δ
3	9,064	10	64,743.70	−67.01
4	9,063	13	64,808.67	−2.04
5	9,062	16	64,799.49	−11.23
6	9,061	19	64,806.76	−3.96
7	9,060	22	64,806.40	−4.31
8	9,059	25	64,796.87	−13.85
9	**9,058**	**28**	**64,810.72**	**0.00**
10	9,057	31	64,798.14	−12.58

The surface is remarkably flat. The runner-up $p = 4$ is only 2.04 log-units below the maximum, and $p = 6$ and $p = 7$ are within 4.5; only $p = 3$ is decisively rejected. In Bayesian model-comparison terms the data are close to indifferent across $p \in \{4, \ldots, 10\}$.

A separate lag-grid search bears on the persistence of the credit-spread equation discussed in Section A.1. Extending the candidate grid to $p \in \{9, 12, 15, 18, 22, 25\}$ finds that the marginal likelihood continues to prefer $p = 9$, declining monotonically thereafter and by 274 log-units at $p = 25$, while residual autocorrelation at lags beyond p is rejected for CSPREAD at *every* candidate order. The persistence is a property of the data source and not a symptom of an under-specified conditional mean.

F.2 Reduced-form and rotation diagnostics

Linear serial correlation is mild, rejecting only for v_1 and only marginally. Volatility clustering rejects overwhelmingly for all three, with ARCH–LM statistics in the hundreds to thousands. This does not bias the rotation, but it invalidates posterior bands that treat $\widehat{K}$ as exact, which is why inference on the eigenvalues is based on the block bootstrap.

Table F.2: Breusch–Godfrey serial correlation test, posterior mean residuals, BVAR($p = 9$). H_0: no serial correlation to order 10.

Variable	BG(10)	p-value	Decision
SP500	33.61	0.0002	Reject H_0
VIX	15.39	0.1185	Fail to reject H_0
CSPREAD	44.79	0.0000	Reject H_0

Table F.3: Serial independence of the identified structural shocks. LB(v): Ljung–Box on v_{kt}; LB(v^2): Ljung–Box on v_{kt}^2; ARCH–LM: Engle test with 10 lags. p-values in parentheses.

Shock	LB(v)	(p)	LB(v^2)	(p)	ARCH–LM	(p)
v_1	20.68	(0.023)	1038.3	(0.000)	574.4	(0.000)
v_2	11.29	(0.335)	8643.0	(0.000)	2346.2	(0.000)
v_3	6.08	(0.809)	495.9	(0.000)	306.2	(0.000)

Table F.4: The non-SV-robust panel of the contemporaneous independence diagnostics, co-kurtosis $\mathbb{E}[v_i^2 v_j^2]$ with $H_0 = 1$, moving-block bootstrap 95% intervals. This panel is *expected* to reject under conditionally independent shocks driven by a common volatility factor, and the magnitude of the rejection estimates Var(ω).

i	j	Estimate	Boot. SE	95% CI	
v_1	v_2	5.299	1.862	[2.187, 9.521]	reject
v_1	v_3	1.498	0.288	[1.032, 2.151]	reject
v_2	v_3	2.938	0.714	[1.767, 4.534]	reject

Table F.5: Reference rotation and posterior sign pattern. Left: P_{ref} at the posterior mean of Σ_u, rows are pre-whitened system variables and columns are structural shocks ordered by λ_i. Right: posterior sign agreement of $B^{(0)}$, where $+$ and $-$ denote agreement of at least 84% and $-?$ agreement below it.

P_{ref}				Sign pattern of $B^{(0)}$			
	v_1	v_2	v_3		v_1	v_2	v_3
SP500	−0.274	0.728	−0.628	SP500	−	+	−
VIX	−0.178	0.604	0.777	VIX	+	−	+
CSPREAD	0.945	0.325	−0.036	CSPREAD	+	+	−?

Eight of the nine signs are robust at the 84% threshold. The single uncertain entry is the credit-spread response to the volatility shock, which is economically sensible: that entry is essentially zero in magnitude, at -0.0001, so its sign is not a meaningful object.

Table F.6: Successive orthogonalisation of the innovations. Correlation matrices at the three stages: the reduced-form residuals, the pre-whitened residuals, and the identified structural shocks. At the reference identification $\max|\mathrm{Corr}(V_{\mathrm{ref}}) - I_3| = 3.45 \times 10^{-4}$.

$\mathrm{Corr}(u_t)$				$\mathrm{Corr}(\boldsymbol{\iota}_t)$				$\mathrm{Corr}(V_{\mathrm{ref}})$			
	SP	VIX	CS		SP	VIX	CS		v_1	v_2	v_3
SP	1.000	−0.719	−0.030	SP	1.0000	0.0004	0.0002	v_1	1.0000	0.0002	−0.0001
VIX	−0.719	1.000	0.032	VIX	0.0004	1.0000	−0.0000	v_2	0.0002	1.0000	0.0003
CS	−0.030	0.032	1.000	CS	0.0002	−0.0000	1.0000	v_3	−0.0001	0.0003	1.0000

The comparison in Table F.6 is the point of the exercise. The reduced-form innovations to equity and implied volatility have a correlation of -0.719, which is the leverage effect in raw form. Whitening reduces it to 4×10^{-4}, and the rotation, being orthogonal, leaves it there. What this shows is what the rotation does *not* do: it does not remove linear dependence, because whitening has already removed it, so whatever the rotation contributes must come from higher moments, which is the premise of the identification. It also places the leverage-effect violation correctly, in the third and fourth moments of the v_2–v_3 pair and not in their covariance, which is zero to four decimals.

F.3 Robustness of the rotation: subsamples and differences

Two objections apply to any identification that consumes fourth moments, and neither is answered by a distributional statistic such as the Kim–White kurtosis. The first is that the rotation may be an artefact of two crisis episodes. The second is that the conditional mean is specified in levels, whereas what the identification consumes are the residuals, for which the Sims et al. (1990) argument provides no analogue. Both are addressed by re-running the reference identification on modified samples and measuring the angle between the resulting rotation columns and the baseline.

Where the fourth moment lives. The premise should be stated before the results, because it is what makes them informative. The tail

mass is extraordinarily concentrated: the 85 trading days of October–November 2008 and March–April 2020, 0.9 per cent of the sample, carry 61 per cent of the fourth-moment content of the pre-whitened residuals, and the ten largest realisations of v_1 alone carry 47 per cent. An estimator built on excess kurtosis is therefore, by construction, an estimator that reads those episodes. Removing both windows removes about two-thirds of the identifying information, and the rotation does not survive it, which is a statement about how much of the fourth moment the crises carry and not a defect of the estimator. The informative exercise is not whether the rotation survives the deletion of the information that identifies it, but whether it rests on more than one episode.

Table F.7: Reference identification re-run on modified samples. The VAR is estimated on the full sample so that the conditional mean is held fixed; the listed observations are removed from the fourth-moment tensor, which isolates the effect on the identification rather than on the dynamics. "Angle" is the angle between the leading column of P and its baseline counterpart, in degrees, taken up to sign.

Sample	T	d_1	κ_1	Angle	Anchors
Baseline, full sample	9,058	42.77	37.97	—	CSPREAD, SP500, VIX
Panel A: one crisis removed					
Excluding 2008	8,805	34.44	31.74	14.7	CSPREAD, SP500, VIX
Excluding 2020	8,805	29.34	24.46	16.8	CSPREAD, CSPREAD, VIX
Panel B: one crisis only					
2007–2009 only	756	26.42	24.21	14.0	CSPREAD, VIX, VIX
2019–2021 only	757	49.94	45.85	2.9	CSPREAD, SP500, VIX

The two windows recover the same object. The leading direction estimated on 2007–2009 is $(-0.494, -0.110, 0.863)$ and the one estimated on 2019–2021 is $(-0.227, -0.171, 0.959)$. The angle between them is 16.7°, and both anchor on CSPREAD with a leading excess kurtosis of 24.2 and 45.9 respectively. Neither window is informed by the other, and neither is informed by the baseline. This is a replication and not a robustness check in the usual sense: the credit direction is recovered twice, from disjoint samples separated by twelve years, using nothing but the fourth moments of three daily prices. Panel A places the same point differently. With 2008 deleted the identification still finds the credit direction to within 15°, because 2020 remains; with 2020 deleted it still finds it to within 17°, because 2008 remains. The labels degrade in the second case, where two shocks anchor on CSPREAD, which is

what one expects when the sample retains one tail episode instead of two.

Table F.8: Reference identification on the first-differenced system. Angles are between each column of P and its baseline counterpart, in degrees, up to sign.

Specification	T	d_i	κ_i	Angles
Levels, $p = 9$ (baseline)	9,058	42.77, 18.64, 11.13	37.97, 12.41, 8.71	—
Differences, $p = 9$	9,057	42.58, 18.10, 11.20	37.75, 11.78, 8.60	0.7, 1.0, 1.1
Differences, $p = 5$	9,061	46.28, 19.47, 10.96	41.26, 13.15, 8.61	0.9, 0.8, 1.2
Differences, $p = 1$	9,065	53.12, 20.37, 10.54	47.74, 13.83, 8.25	2.7, 3.1, 2.0

The identification is invariant to the choice. At the baseline lag order the three rotation columns move by about one degree, the eigenvalues by less than three per cent, and the anchor variables not at all. Even at $p = 1$, where the conditional mean is deliberately impoverished, the columns move by three degrees. Whatever the levels specification does to the conditional mean, it does not propagate into the fourth moments of the residuals, so the concern that motivates the differenced specification for daily financial data does not bind here. Both exercises are computed at the reference identification and are not propagated through the posterior or the block bootstrap; they measure the sensitivity of the point-identified rotation to the sample and to the specification, which is what the two objections concern.

F.4 Narrative check on the monthly aggregates

The separation survives aggregation intact. The credit column is the post-Lehman funding freeze and its aftermath together with the COVID credit seizure; the volatility column is LTCM and the Russian default, September 2001, September 2008, the Flash Crash, the Chinese devaluation and the onset of COVID; the equity column is the dot-com peak of March 2000, the trough of the ensuing bear market in the summer and autumn of 2002, and the capitulation and turn of early 2009, which are the equity market's own inflection points and not the months in which either of the other shocks is largest. No month appears in two columns, which is the empirical counterpart of the claim that the signed mean is the filter preserving the object identified upstream.

Table F.9: The six largest absolute realisations of the signed monthly mean $Z_m^{(k)}$ of (C.1). Values are signed and follow the bridge convention of Section C.3, under which a positive entry is the adverse direction of that shock. The lower panel summarises the distribution of the aggregate over all 432 months.

v_1 (credit)		v_2 (equity)		v_3 (volatility)	
Month	$Z_m^{(1)}$	Month	$Z_m^{(2)}$	Month	$Z_m^{(3)}$
2008-10	+1.81	2009-03	−0.81	2015-08	+0.78
2020-03	+1.72	2009-02	+0.81	2020-02	+0.72
2008-11	+1.22	2009-01	+0.71	2008-09	+0.64
2020-04	−0.94	2000-03	−0.70	1998-08	+0.63
2009-06	−0.84	2002-09	+0.59	2001-09	+0.60
2009-07	−0.70	2002-07	+0.55	2010-05	+0.59
s.d.	0.225	s.d.	0.209	s.d.	0.202
skewness	+2.16	skewness	−0.07	skewness	+0.57
exc. kurt.	19.22	exc. kurt.	1.24	exc. kurt.	1.00

Table F.10: Day-zero structural contributions to the forecast error $\hat{u}_t$ at three crisis dates, original units. The five-day cumulative panel is in the paper.

Event	Variable	v_1	v_2	v_3	Total
Lehman	SP500	−0.0121	−0.0156	−0.0197	−0.0474
	VIX	0.0192	0.0132	0.1848	0.2172
	CSPREAD	0.0755	−0.0120	−0.0003	0.0632
COVID	SP500	−0.0093	−0.0781	−0.0332	−0.1206
	VIX	0.0148	0.0663	0.3111	0.3923
	CSPREAD	0.0583	−0.0603	−0.0005	−0.0025
SVB	SP500	−0.0043	−0.0006	−0.0106	−0.0154
	VIX	0.0068	0.0005	0.0989	0.1062
	CSPREAD	0.0269	−0.0005	−0.0002	0.0262

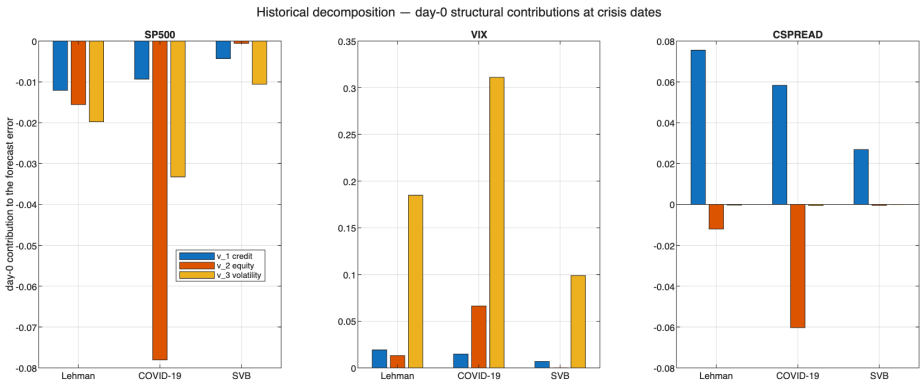


Figure F.1: Structural contributions at the three crisis dates, one bar per shock and variable.

F.5 Historical decomposition: day-zero panel

F.6 External validation: full panels and the filter

The paper reports the signed and the amplitude designs on a reduced set of indices. Table F.11 gives all three designs on all five indices. Design A matches the intensity aggregate against the *level* of the index and asks whether the shock is more violent when measured uncertainty is high; both series are persistent, which makes that panel descriptive and not inferential. Design B matches the signed mean against the AR(6) innovation $\hat{\eta}$ and carries the inferential weight. Design C asks Design A's amplitude question with the persistence removed from both sides.

Table F.11: Full correlation table. Pearson coefficients; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Each coefficient is computed on the intersection of the index's own support with the monthly shock sample, which is why coefficients of similar size can carry different stars. Signs in Panel B follow the bridge convention; Panels A and C are invariant to it, being even functions of the shock.

	v_1 (credit)	v_2 (equity)	v_3 (volatility)
Panel A: index level against intensity (descriptive)			
US_EPU	**0.348*****	0.284***	0.300***
GEPU_current	0.259***	0.132**	**0.306*****
GEPU_ppp	0.260***	0.120**	0.299***
GPR	0.044	0.054	0.006
GPRC_USA	0.046	0.055	0.003
Panel B: index innovation against signed mean (inferential)			
US_EPU	0.088*	0.065	**0.238*****
GEPU_current	0.086	0.131**	**0.204*****
GEPU_ppp	0.087	0.127**	0.172***
GPR	0.007	**0.148*****	0.068
GPRC_USA	−0.006	**0.136*****	0.074
Panel C: $\lvert\hat{\eta}\rvert$ against intensity (amplitude, both sides filtered)			
US_EPU	0.217***	0.225***	0.199***
GEPU_current	0.210***	0.187***	0.200***
GEPU_ppp	0.219***	0.173***	0.194***
GPR	0.046	0.054	0.117**
GPRC_USA	0.047	0.045	0.124**

Table F.12 sets the two filters against each other, and the comparison is itself the evidence that neither is driving the result. The estimated persistence splits the indices into two families. The EPU family is close

Table F.12: The two filters of the index in the signed design, side by side. $\hat{\rho} = \sum_j \rho_j$ is the persistence estimated by the AR(6) on the log level, against the unit value $\Delta \log$ imposes.

		v_1 (credit)		v_2 (equity)		v_3 (volatility)	
Index	$\hat{\rho}$	$\hat{\eta}$	$\Delta \log$	$\hat{\eta}$	$\Delta \log$	$\hat{\eta}$	$\Delta \log$
US_EPU	0.953	0.088	0.041	0.065	0.096	0.238	0.234
GEPU_current	0.973	0.086	0.061	0.131	0.129	0.204	0.181
GEPU_ppp	0.978	0.087	0.063	0.127	0.120	0.172	0.151
GPR	0.854	0.007	0.012	0.148	0.141	0.068	0.088
GPRC_USA	0.877	−0.006	0.001	0.136	0.134	0.074	0.090

to integrated, at 0.953 to 0.978, so the unit root that $\Delta \log$ imposes is nearly correct and the two columns agree to within a few hundredths. The GPR family is genuinely mean-reverting, at 0.854 and 0.877, so $\Delta \log$ over-differences and the columns separate. The filters agree where theory says they should and differ where it says they should, which is a stronger statement than either column would support alone, and where they differ the estimated-persistence column is the correct one on statistical grounds, since it alone pairs two innovations. Ten of the fifteen coefficients rise when the persistence is estimated, the two largest movements being the credit shock against US_EPU, which doubles from 0.041 to 0.088, and the geopolitical null for the volatility shock, which falls below significance. That is the single place where the filter changes a conclusion, and it changes it towards the null.

F.7 Serial correlation of all four monthly aggregates

Conditional heteroskedasticity is confirmed for the instrument itself: the squared signed means are rejected for v_1 and v_2, with $\rho(1) = 0.38$ and 0.36. This is in addition to the calendar effect $\mathrm{Var}(Z_m) = \sigma^2/D_m$, and together they are the empirical justification for the block-bootstrap inference of Section C.7 in place of a Gaussian posterior band. For v_3 the squared mean is not rejected, consistent with its being the least leptokurtic of the three shocks.

Table F.13: Serial correlation diagnostics on the four monthly aggregates of each daily shock, 432 months. $\rho(1)$ is the first-order autocorrelation and LB(12) the Ljung–Box statistic on twelve monthly lags, distributed $\chi^2(12)$ under the null. The signed mean is the instrument; the other three are reported for comparison.

Shock	Aggregate	$\rho(1)$	LB(12)	p-value	Reading
v_1	mean_k	0.169	34.4	0.0006	autocorrelated
	mean_k^2	0.384	79.6	0.0000	cond. heterosk.
	abs_k	0.674	615.7	0.0000	not an instrument
	sq_k	0.462	143.0	0.0000	not an instrument
v_2	mean_k	−0.001	24.1	0.0198	rejected at 5%
	mean_k^2	0.359	103.6	0.0000	cond. heterosk.
	abs_k	0.626	484.0	0.0000	not an instrument
	sq_k	0.477	172.9	0.0000	not an instrument
v_3	mean_k	−0.038	19.8	0.0715	marginal
	mean_k^2	0.151	18.0	0.1160	—
	abs_k	0.292	160.0	0.0000	not an instrument
	sq_k	0.212	89.6	0.0000	not an instrument

Table F.14: Shock predictability from lagged macroeconomic information, $T = 414$. Panel A: joint significance of the eight regressors, HC1-robust Wald statistic reported as W/q with $\chi^2(8)$ p-values. Panel B: robust t-statistics on individual regressors for v_1; * denotes $|t| > 1.96$. Regressors are standardised, so the t-statistics are comparable across variables.

Panel A: joint test

Shock	R^2	W/q	p-value	Result
v_1 (credit)	0.102	2.20	0.0248	**Reject** at 5%, not at 1%
v_2 (equity)	0.026	0.87	0.5435	Do not reject
v_3 (volatility)	0.017	1.92	0.0524	Do not reject

Panel B: robust t-statistics, shock v_1

$\Delta \ln \text{INDPRO}(-1)$	−0.85	$\Delta \ln \text{INDPRO}(-2)$	-2.62^*
$\Delta\text{UNRATE}(-1)$	−1.47	$\Delta\text{UNRATE}(-2)$	-2.03^*
$\Delta \ln \text{PCEPI}(-1)$	−0.09	$\Delta \ln \text{PCEPI}(-2)$	+0.26
$\text{EBP}(-1)$	$+1.99^*$	$\text{EBP}(-2)$	-2.02^*

F.8 Lag robustness at the monthly stage

Section C.4 argues that under the GLP prior the lag order may be fixed generously because the shrinkage, and not the truncation, does the work. The argument is verified by re-estimating the entire system at $p_m \in \{12, 15, 18\}$ and correlating the resulting impulse responses with the baseline, after removing the normalisation so that shape rather than scale is compared.

The result is unambiguous in both directions. Moving from $p_m = 15$ to $p_m = 18$ changes nothing, with correlations between 0.987 and 0.999 on every (shock, variable) pair, which means the prior has already shrunk the additional lags to irrelevance, exactly as the $1/s^2$ decay is supposed to do. Moving to $p_m = 12$ does change the responses materially, with correlations as low as 0.345 and one sign reversal. The specification therefore sits on the plateau and not at an arbitrary point on a slope, and a shorter lag order truncates dynamics the data contain, which is the asymmetry of errors invoked in Section C.4. That the responses peak between $h = 2$ and $h = 25$ makes the finding intuitive: a twelve-lag conditional mean cannot represent dynamics that are still building at the end of its own window.

F.9 Impulse responses: the complete record

Tables F.15–F.17 report every macroeconomic impulse response underlying Section 5 of the paper: all ten variables, all three shocks, at six horizons, with the moving-block bootstrap interval at each. The paper tabulates the peak of each of these paths and three variables over the horizon; what these tables add is the shape in between, which is where the persistence of the credit shock and the recovery of the volatility shock can be read directly. Figures F.3 and F.4 then give the equity and volatility panels in full, the credit panel being in the paper.

F.10 Distributional diagnostics

Both sets of QQ-plots show the same qualitative departure, a pronounced S-shaped deviation in both tails, which is what the kurtosis and Hill statistics report numerically. Beyond illustration, two features bear on the identification itself. The departure is present in *both* tails and in the bulk and is not confined to a handful of extreme days,

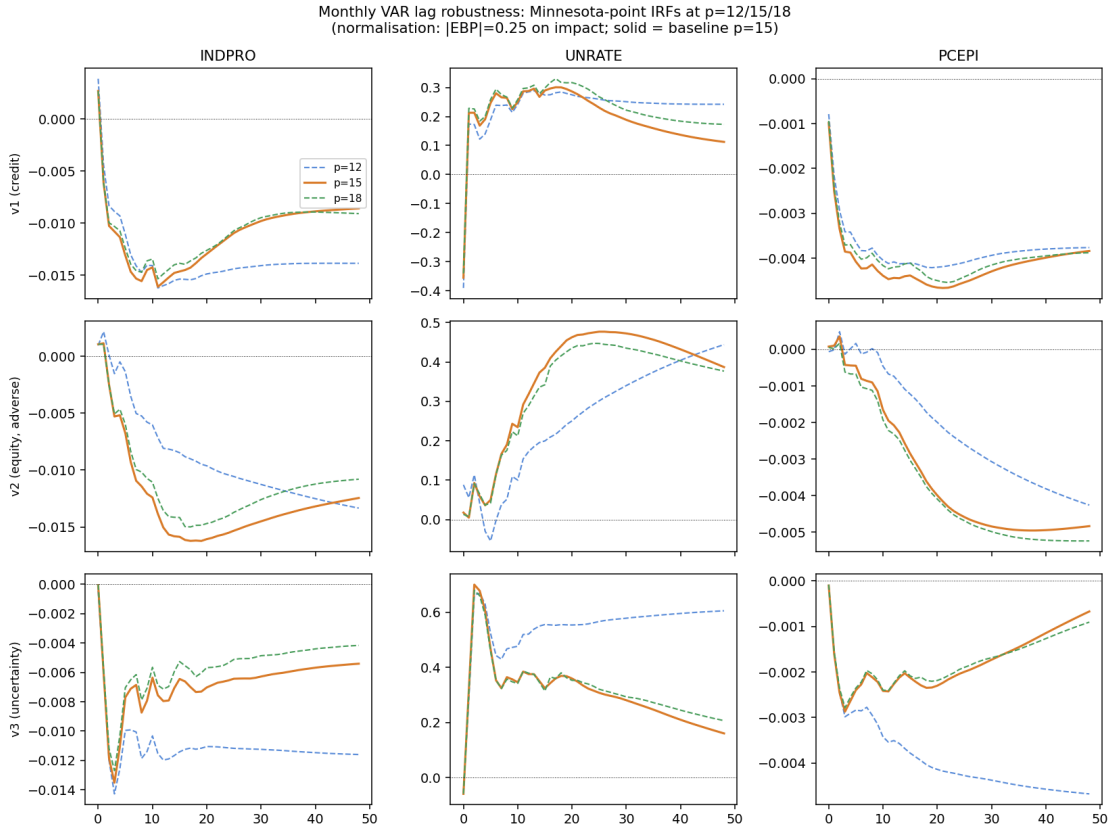


Figure F.2: Impulse responses at $p_m = 12$, 15 and 18.

Table F.15: Full macroeconomic impulse responses to the credit shock v_1. Minnesota point estimate with [16, 84] moving-block bootstrap percentiles on the same estimator, per shock moving the excess bond premium by 25 basis points on impact. Log-level variables in percent; UNRATE, FFR, RAI_BEX, UNC_BEX and EBP in percentage points.

Variable	$h = 0$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 48$
INDPRO	+0.26 [−0.10, +0.31]	−1.48 [−1.52, −1.11]	−1.57 [−1.72, −1.09]	−1.12 [−1.50, −0.59]	−0.90 [−1.31, −0.47]	−0.85 [−1.21, −0.44]
UNRATE	−0.36 [−0.41, −0.03]	+0.28 [+0.15, +0.42]	+0.28 [+0.18, +0.40]	+0.23 [+0.11, +0.41]	+0.15 [+0.05, +0.34]	+0.11 [+0.03, +0.28]
HOUST	+0.62 [−0.50, +1.47]	−5.18 [−6.10, −2.47]	−3.85 [−4.51, −1.29]	−2.52 [−3.54, +0.04]	−2.14 [−3.25, +0.36]	−2.08 [−3.10, +0.46]
CSHPI	−0.03 [−0.05, −0.00]	−0.56 [−0.72, −0.32]	−0.52 [−0.89, −0.11]	−0.37 [−1.00, +0.30]	−0.26 [−0.98, +0.59]	−0.23 [−0.96, +0.72]
PCEPI	−0.10 [−0.13, −0.07]	−0.42 [−0.50, −0.27]	−0.44 [−0.52, −0.27]	−0.45 [−0.57, −0.26]	−0.40 [−0.55, −0.22]	−0.38 [−0.52, −0.19]
NRCREDIT	−0.02 [−0.10, +0.03]	−0.22 [−0.47, −0.03]	−0.42 [−0.73, −0.16]	−0.64 [−1.00, −0.20]	−0.48 [−0.92, −0.03]	−0.36 [−0.86, +0.08]
FFR	−0.05 [−0.07, −0.02]	−0.04 [−0.10, +0.10]	−0.01 [−0.15, +0.12]	+0.01 [−0.20, +0.15]	+0.05 [−0.17, +0.18]	+0.09 [−0.13, +0.21]
RAI_BEX	+0.46 [+0.37, +0.55]	+0.14 [+0.04, +0.19]	+0.07 [+0.02, +0.11]	+0.02 [−0.00, +0.07]	+0.02 [+0.00, +0.07]	+0.03 [+0.00, +0.06]
UNC_BEX	+0.27 [+0.21, +0.36]	+0.14 [+0.09, +0.17]	+0.04 [+0.00, +0.08]	−0.03 [−0.05, +0.02]	−0.01 [−0.03, +0.02]	−0.00 [−0.02, +0.02]
EBP	+0.25 [+0.25, +0.25]	+0.20 [+0.14, +0.24]	+0.00 [−0.05, +0.05]	−0.04 [−0.06, +0.00]	−0.02 [−0.03, +0.00]	−0.01 [−0.02, +0.00]

Table F.16: Full macroeconomic impulse responses to the equity shock v_2. Minnesota point estimate with $[16, 84]$ moving-block bootstrap percentiles on the same estimator, per shock moving the excess bond premium by 25 basis points on impact. Log-level variables in percent; UNRATE, FFR, RAI_BEX, UNC_BEX and EBP in percentage points.

Variable	$h = 0$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 48$
INDPRO	$+0.14\,[-0.04, +0.46]$	$-0.84\,[-1.30, -0.33]$	$-1.40\,[-2.03, -0.67]$	$-1.44\,[-2.26, -0.57]$	$-1.27\,[-2.10, -0.49]$	$-1.16\,[-1.90, -0.41]$
UNRATE	$+0.02\,[-0.07, +0.09]$	$+0.10\,[-0.10, +0.28]$	$+0.29\,[+0.11, +0.50]$	$+0.45\,[+0.19, +0.69]$	$+0.43\,[+0.18, +0.68]$	$+0.37\,[+0.15, +0.60]$
HOUST	$-2.51\,[-5.88, -1.18]$	$-5.92\,[-8.91, -2.61]$	$-6.29\,[-8.99, -2.28]$	$-5.23\,[-7.82, -1.02]$	$-4.52\,[-7.21, -0.41]$	$-3.99\,[-6.73, +0.06]$
CSHPI	$-0.12\,[-0.26, -0.05]$	$-1.43\,[-2.59, -0.63]$	$-2.23\,[-3.93, -0.95]$	$-3.03\,[-5.14, -1.05]$	$-3.21\,[-5.42, -0.87]$	$-3.21\,[-5.34, -0.67]$
PCEPI	$+0.01\,[-0.04, +0.07]$	$-0.07\,[-0.22, +0.06]$	$-0.19\,[-0.43, -0.00]$	$-0.43\,[-0.79, -0.11]$	$-0.48\,[-0.88, -0.11]$	$-0.47\,[-0.88, -0.09]$
NRCREDIT	$-0.04\,[-0.24, +0.06]$	$+0.31\,[-0.12, +0.72]$	$+0.31\,[-0.26, +0.78]$	$-0.06\,[-0.80, +0.53]$	$+0.00\,[-0.97, +0.56]$	$+0.20\,[-0.90, +0.72]$
FFR	$+0.13\,[+0.07, +0.23]$	$+0.28\,[+0.04, +0.51]$	$+0.28\,[-0.06, +0.50]$	$+0.07\,[-0.29, +0.26]$	$-0.05\,[-0.39, +0.18]$	$-0.08\,[-0.37, +0.15]$
RAI_BEX	$+0.45\,[+0.32, +0.73]$	$+0.16\,[+0.07, +0.22]$	$+0.16\,[+0.08, +0.25]$	$+0.07\,[+0.02, +0.16]$	$+0.04\,[+0.01, +0.13]$	$+0.03\,[+0.00, +0.10]$
UNC_BEX	$+0.11\,[+0.07, +0.20]$	$+0.23\,[+0.13, +0.32]$	$+0.20\,[+0.09, +0.30]$	$+0.10\,[+0.03, +0.17]$	$+0.06\,[+0.00, +0.11]$	$+0.05\,[-0.01, +0.09]$
EBP	$+0.25\,[+0.25, +0.25]$	$+0.23\,[+0.14, +0.28]$	$+0.13\,[+0.03, +0.20]$	$+0.04\,[-0.02, +0.08]$	$+0.01\,[-0.04, +0.03]$	$-0.00\,[-0.03, +0.01]$

Table F.17: Full macroeconomic impulse responses to the volatility shock v_3. Minnesota point estimate with $[16, 84]$ moving-block bootstrap percentiles on the same estimator, per shock moving the excess bond premium by 25 basis points on impact. Log-level variables in percent; UNRATE, FFR, RAI_BEX, UNC_BEX and EBP in percentage points.

Variable	$h = 0$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 48$
INDPRO	$-0.04\,[-0.19, +0.10]$	$-0.71\,[-0.89, -0.28]$	$-0.79\,[-1.16, -0.33]$	$-0.62\,[-1.09, -0.20]$	$-0.56\,[-0.95, -0.08]$	$-0.51\,[-0.84, +0.05]$
UNRATE	$-0.05\,[-0.09, +0.05]$	$+0.36\,[+0.11, +0.51]$	$+0.38\,[+0.18, +0.53]$	$+0.31\,[+0.18, +0.49]$	$+0.24\,[+0.13, +0.41]$	$+0.16\,[+0.07, +0.33]$
HOUST	$+1.17\,[+0.07, +2.37]$	$-3.43\,[-4.42, -0.77]$	$-3.29\,[-4.24, -0.66]$	$-1.57\,[-3.19, +0.63]$	$-1.06\,[-2.60, +1.41]$	$-0.75\,[-2.22, +1.97]$
CSHPI	$-0.10\,[-0.15, -0.06]$	$-0.90\,[-1.28, -0.51]$	$-0.77\,[-1.52, -0.18]$	$-0.49\,[-1.75, +0.27]$	$-0.14\,[-1.64, +0.77]$	$+0.16\,[-1.50, +1.18]$
PCEPI	$-0.01\,[-0.04, +0.02]$	$-0.22\,[-0.29, -0.11]$	$-0.22\,[-0.30, -0.12]$	$-0.20\,[-0.36, -0.08]$	$-0.14\,[-0.34, +0.00]$	$-0.07\,[-0.30, +0.09]$
NRCREDIT	$-0.16\,[-0.33, -0.10]$	$-0.07\,[-0.36, +0.01]$	$-0.05\,[-0.50, +0.08]$	$+0.13\,[-0.59, +0.25]$	$+0.36\,[-0.55, +0.49]$	$+0.52\,[-0.45, +0.72]$
FFR	$-0.04\,[-0.06, -0.01]$	$-0.17\,[-0.27, -0.07]$	$-0.18\,[-0.36, -0.07]$	$-0.21\,[-0.43, -0.10]$	$-0.16\,[-0.40, -0.05]$	$-0.09\,[-0.34, +0.02]$
RAI_BEX	$+0.89\,[+0.73, +1.07]$	$+0.13\,[+0.05, +0.17]$	$+0.19\,[+0.13, +0.25]$	$+0.14\,[+0.10, +0.20]$	$+0.11\,[+0.07, +0.16]$	$+0.08\,[+0.06, +0.14]$
UNC_BEX	$+0.39\,[+0.31, +0.47]$	$+0.18\,[+0.10, +0.23]$	$+0.15\,[+0.08, +0.21]$	$+0.12\,[+0.05, +0.16]$	$+0.10\,[+0.04, +0.13]$	$+0.09\,[+0.04, +0.11]$
EBP	$+0.25\,[+0.25, +0.25]$	$+0.08\,[+0.02, +0.12]$	$+0.01\,[-0.06, +0.07]$	$-0.00\,[-0.04, +0.02]$	$-0.02\,[-0.04, +0.00]$	$-0.02\,[-0.04, -0.00]$

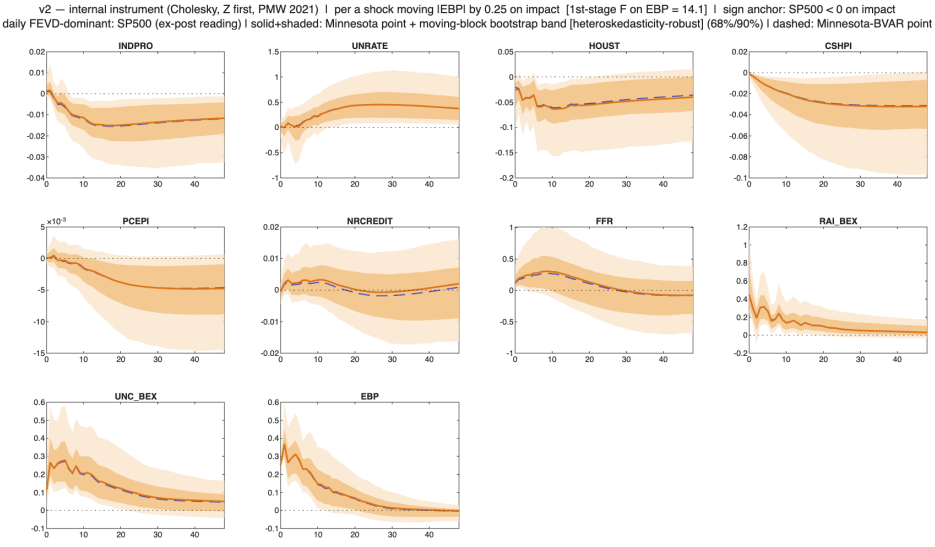


Figure F.3: Macroeconomic impulse responses to the equity shock v_2, all ten variables. Solid line: Minnesota posterior mean; shaded: $[16, 84]$ and $[5, 95]$ moving-block bootstrap percentiles on the same estimator; dashed: Gaussian posterior band.

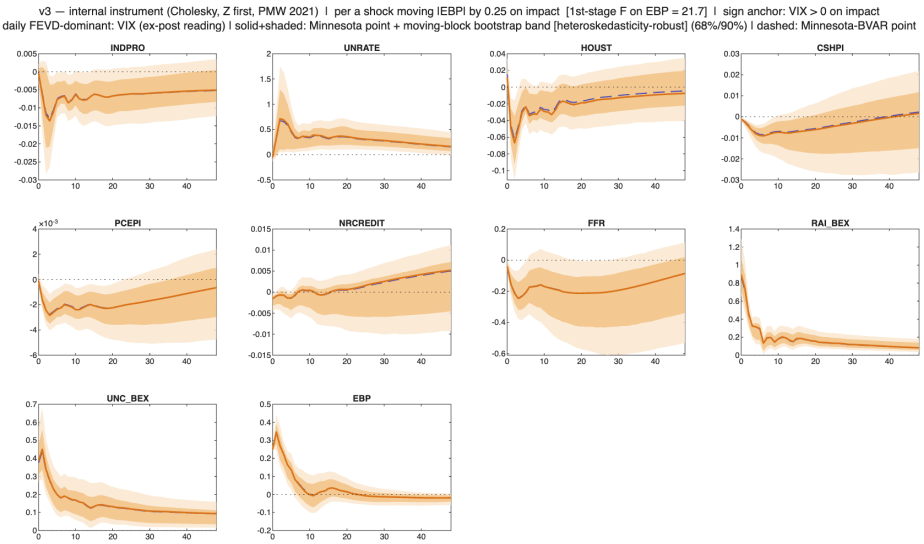


Figure F.4: Macroeconomic impulse responses to the volatility shock v_3, all ten variables. Conventions as in Figure F.3.

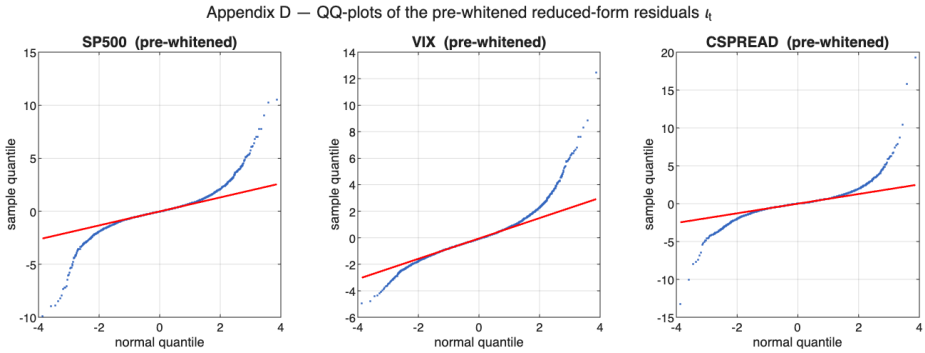


Figure F.5: Normal QQ-plots of the pre-whitened reduced-form residuals $\boldsymbol{\iota}_t$. The reference line is fitted on the interquartile range rather than by least squares, so that it is not dragged by the tails the plot exists to display.

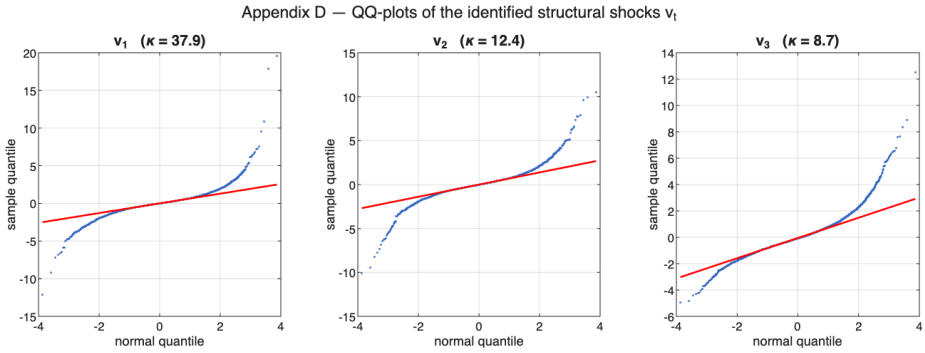


Figure F.6: Normal QQ-plots of the identified structural shocks $\mathbf{v}_t$, each panel labelled by the shock's realised excess kurtosis at the *reference* identification (37.9, 12.4, 8.7). The posterior medians of the same quantities differ from these in the second significant digit.

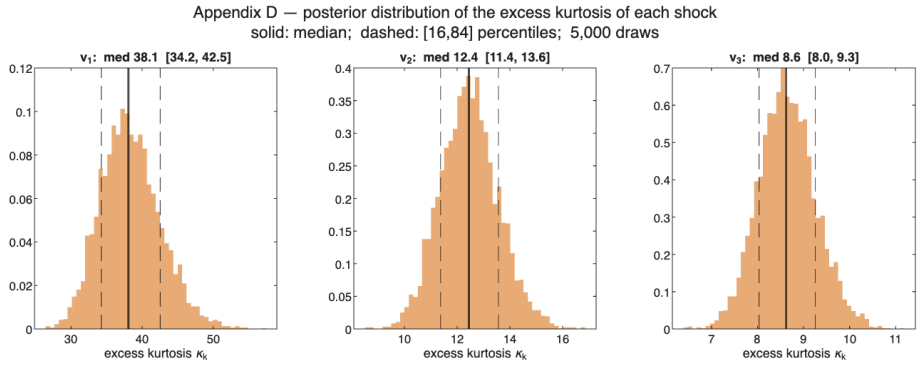


Figure F.7: Posterior distribution of the realised excess kurtosis of each identified shock, over 5,000 draws.

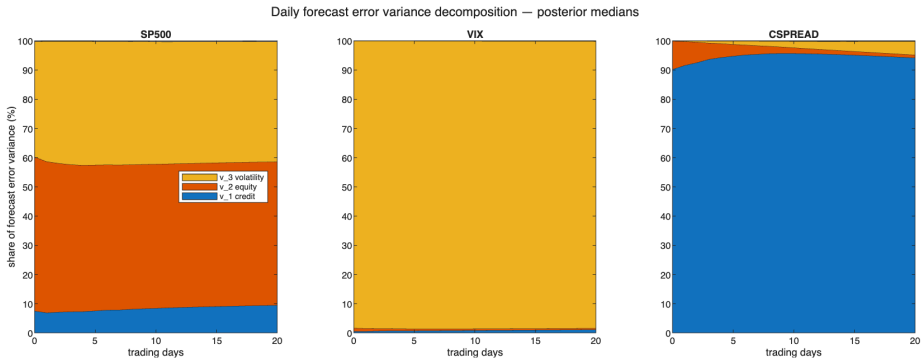


Figure F.8: Daily forecast error variance decomposition, horizon by horizon. Each panel is one system variable; the shaded areas are the shares of its h-step forecast error variance attributable to the three shocks, summing to one at every horizon.

which is the graphical counterpart of the Kim–White result. And the structural shocks are *more* leptokurtic than the residuals from which they are constructed, which is the visual signature of a rotation that concentrates non-Gaussianity into individual components instead of spreading it.

F.11 Reproducibility

The objects in this section are produced by three scripts. The independence diagnostics with bootstrap standard errors come from the block-bootstrap routine on the identified shocks, which writes its estimates, standard errors and intervals to file rather than to the console alone. The correlation matrices, QQ-plots and kurtosis posteriors come from a diagnostics routine that *loads* the stored identification and re-estimates nothing, so running it cannot alter any number reported elsewhere. The reduced-form comparison of the flat and Minnesota priors underlying Section B.1 comes from a separate script run on the same three-variable baseline; its lag-selection output independently reproduces Table F.1, selecting $p = 9$ with the same marginal-likelihood ordering, and its shrinkage diagnostic is reported in Table B.1.

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